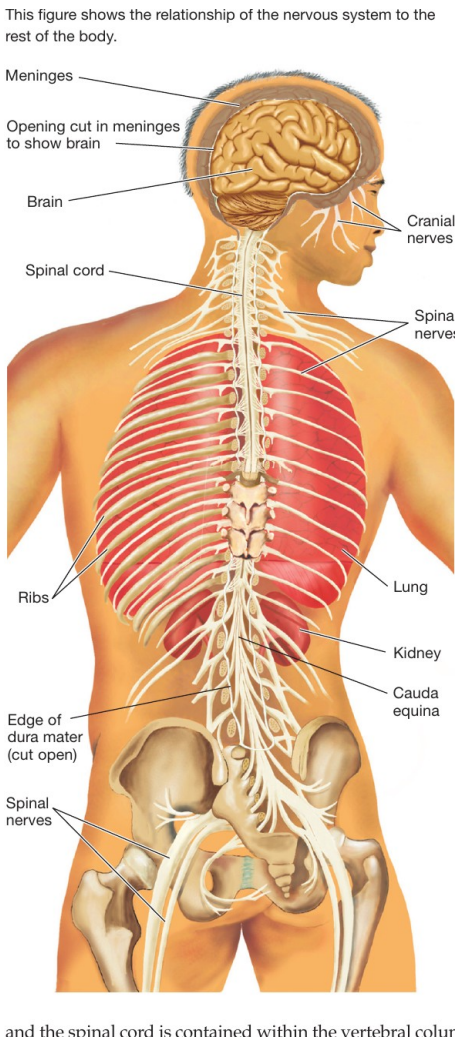


Carlson and Birkett (2022), Fig. 3.1, p. 70.

BEHAVIORAL NEUROSCIENCE 392

Anatomy, Pathways, and Behavior

Mapping the Nervous System: From Brain Structures to Behavior

Fall 2026

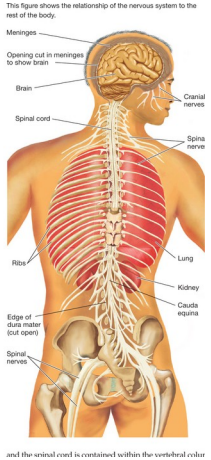
Jordan Lopez, Ph.D.
Professor of Neurobiology & Biomedical Engineering

The nervous system carries signals between brain and body

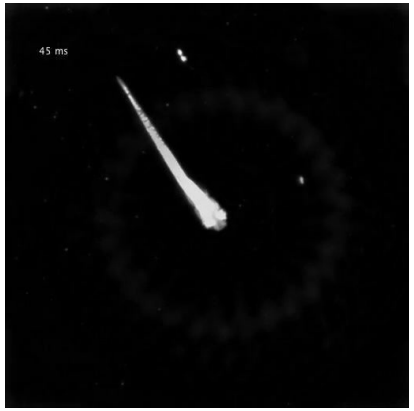
The **central nervous system**, or CNS, consists of the brain and spinal cord. The **peripheral nervous system**, or PNS, includes nerves and ganglia outside those structures. Together, these divisions carry information from sensory receptors, process that information, and send commands to muscles and glands throughout the body.

A **nerve** is a bundle of axons in the peripheral nervous system. A **ganglion** is a collection of neuronal cell bodies outside the brain and spinal cord. These terms identify different kinds of tissue: the bundle carries signals, while the ganglion contains cells that send or receive them.

An animal catching prey must detect the target, turn toward it, and coordinate many muscles. Sensory signals enter central circuits through peripheral pathways, and motor commands return to the body. In the recording, larval zebrafish track prey through coordinated turns and swimming, making cooperation between sensory input and motor output directly visible.



Carlson and Birkett (2022), Fig. 3.1, p. 70.



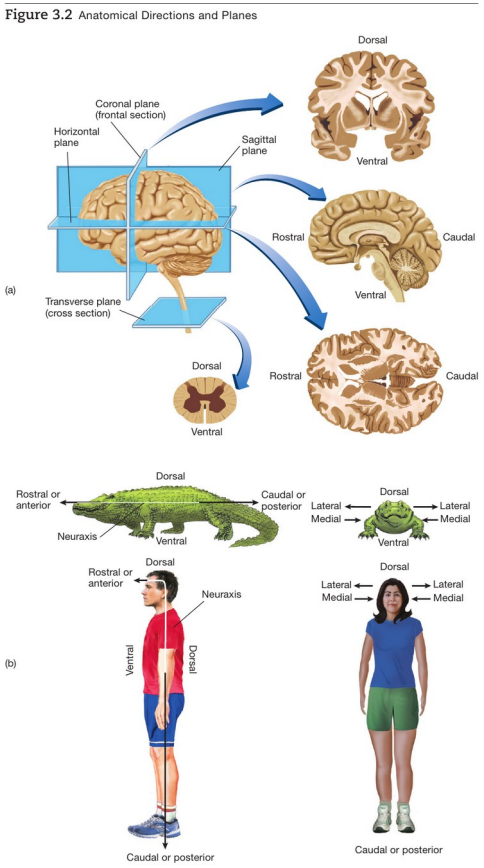
Cong et al. (2017), Video 5. Prey tracking.

Anatomical directions depend on the nervous system's axis

The **neuraxis** is an imaginary line running through the brain and spinal cord. **Rostral** means toward the nose, and **caudal** means toward the tail or lower spinal cord. These directions describe the position of one structure relative to another along the nervous system's main axis.

Dorsal means toward the back, while **ventral** means toward the belly. The human neuraxis bends between the forebrain and spinal cord. Dorsal therefore points toward the top of the forebrain but toward the back of the spinal cord, rather than toward one fixed direction in the room.

Medial means toward the midline, and **lateral** means toward the side. **Ipsilateral** structures lie on the same side; **contralateral** structures lie on opposite sides. A left-sided brain region can send signals to a right-sided muscle, so the side of the source and the side of its target both matter.



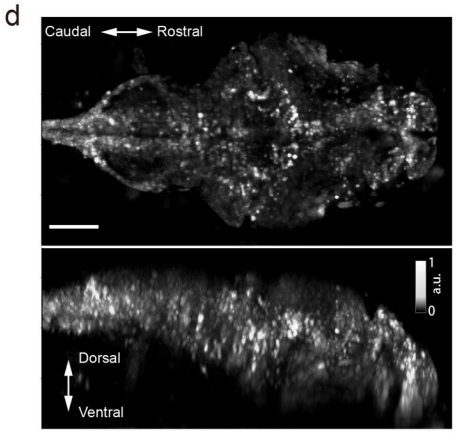
Carlson and Birkett (2022), Fig. 3.2, p. 71.

Imaging reveals activity across an intact larval brain

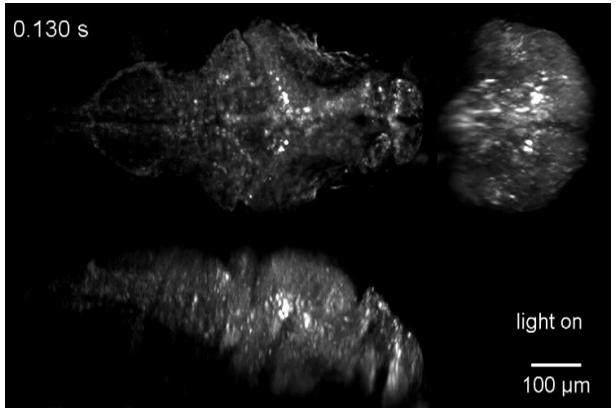
Cong and colleagues developed a microscope that records neural activity across the brain of a larval zebrafish. The neurons produced a fluorescent **calcium indicator**, a protein whose brightness changes with cellular calcium. The preparation allowed researchers to observe activity in many regions while keeping their anatomical positions identifiable.

The investigators delivered a light stimulus while recording the brain. Different locations changed fluorescence over time, making the passage from sensory stimulation to distributed activity observable. A bright region marks calcium changes in the labeled tissue; it does not directly show the electrical voltage of every individual neuron.

Anatomical orientation makes these recordings interpretable. A dorsal view places the two sides of the brain beside each other and separates rostral regions from caudal regions. Comparing activity at known locations helps researchers ask which parts respond to a sensory event and when those responses occur relative to one another.



Cong et al. (2017), Fig. 1d. Brain volume views.



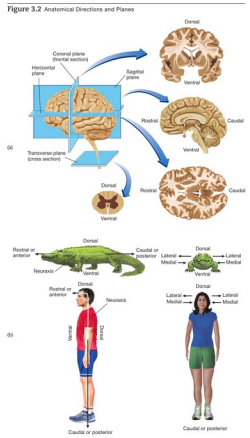
Cong et al. (2017), Video 1. Brain activity during light stimulation.

Brain sections expose different views of the same anatomy

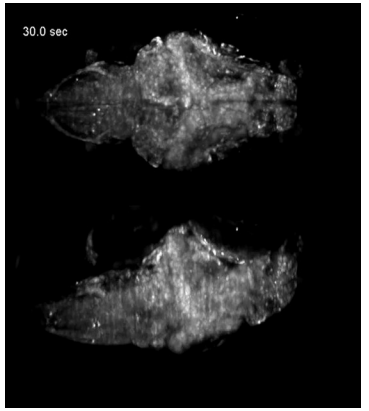
A **coronal section** cuts across the brain from side to side, separating front from back. A **sagittal section** separates left from right. A **horizontal section** separates upper from lower portions. Each plane reveals a different arrangement of the same structures rather than a different kind of brain.

A section through the midline is **midsagittal**. It exposes structures that may be hidden when the brain is viewed from its outer surface, including the corpus callosum and parts of the ventricular system. A section farther to the side passes through a different set of structures within one hemisphere.

Microscopes can also collect images at several depths through intact tissue. The accompanying recording shows spontaneous activity in a larval brain, viewed through an optical imaging method. Identifying the viewing direction and the tissue depth is necessary before comparing a fluorescent region with a structure in a physical section.



Carlson and Birkett (2022), Fig. 3.2, p. 71.



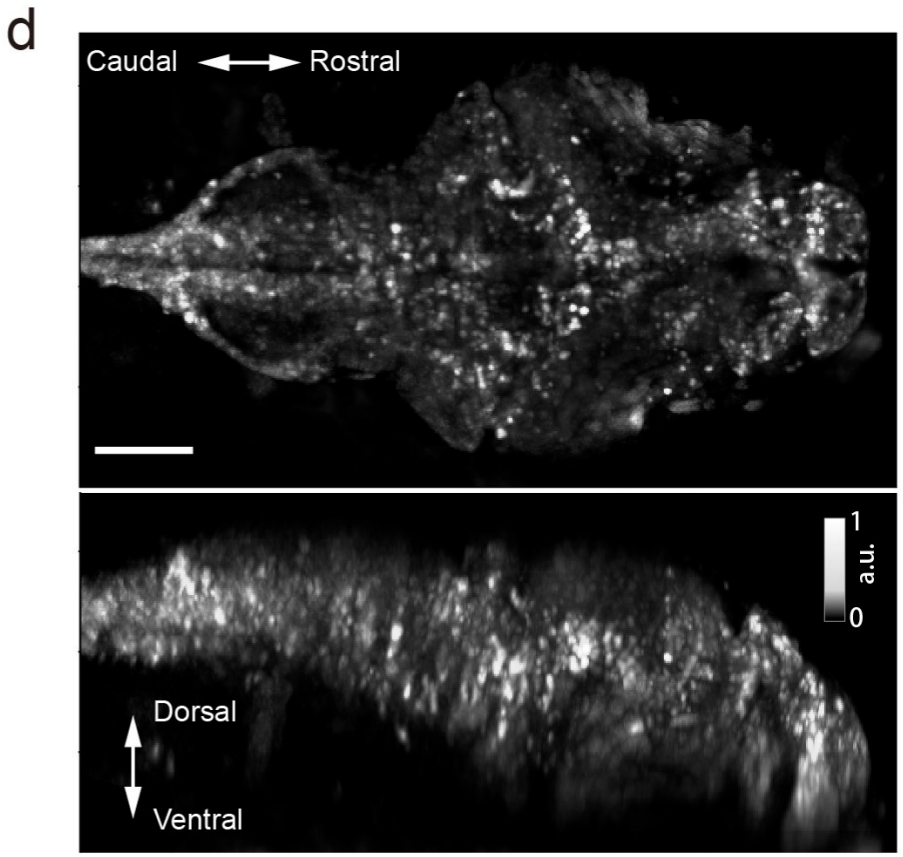
Cong et al. (2017), Video 2. Spontaneous nuclear-labeled brain activity.

Brain imaging preserves location while following activity

The zebrafish imaging system combined information from several optical views to recover activity throughout a volume. A **volume** includes width, height, and depth. Recording a volume therefore samples tissue above and below a single surface, allowing researchers to follow activity across several anatomical regions during the same event.

Cong and colleagues recorded both spontaneous activity and responses to visual stimulation. **Spontaneous activity** occurs without the experimenter delivering the particular stimulus being tested. Comparing these recordings separates ongoing fluctuations from changes that occur around the onset of an external event, while retaining the location of the responding tissue.

The study used fluorescent indicators located either in cell nuclei or throughout the cell body. Nuclear labeling helps separate neighboring cells; broader cellular labeling fills more tissue. The position of the indicator changes what the image emphasizes, even though both approaches use calcium-dependent fluorescence to monitor activity in living neural tissue.



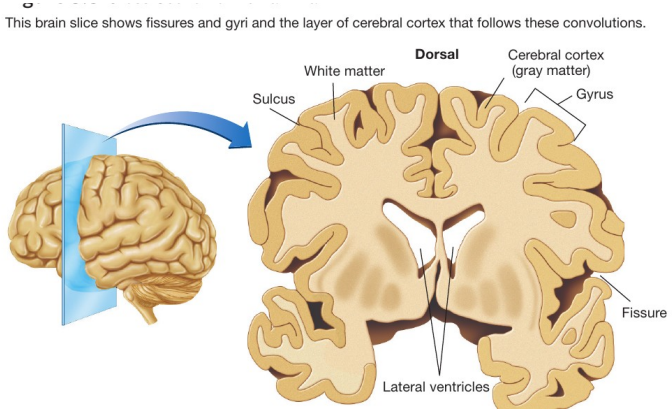
Cong et al. (2017), Fig. 1d. Brain volume views.

Gray and white matter organize cell bodies and axons

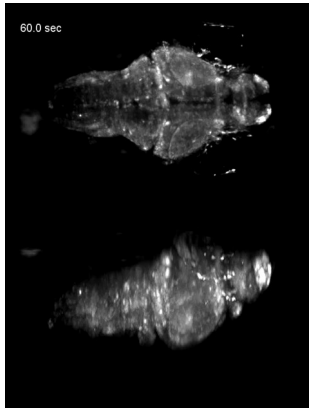
Gray matter contains many neuronal cell bodies, dendrites, and synapses. **White matter** contains many myelinated axons traveling between locations. The color difference reflects tissue composition. Both contain supporting cells and blood vessels, and both participate in communication by placing different parts of the signaling pathway in different locations.

The **cerebral cortex** is a sheet of gray matter covering the cerebral hemispheres. Much of the underlying white matter carries axons between cortical regions and deeper structures. A cross section reveals this arrangement more clearly than a view of the outer surface, where the internal bundles remain hidden.

Fluorescent brain recordings emphasize labeled cells and their activity rather than the natural color of gray and white matter. The accompanying spontaneous-activity recording uses a cellular indicator in larval zebrafish. Interpreting it requires distinguishing the anatomical tissue being imaged from the fluorescence signal used to observe that tissue.



Carlson and Birkett (2022), Fig. 3.8, p. 76.



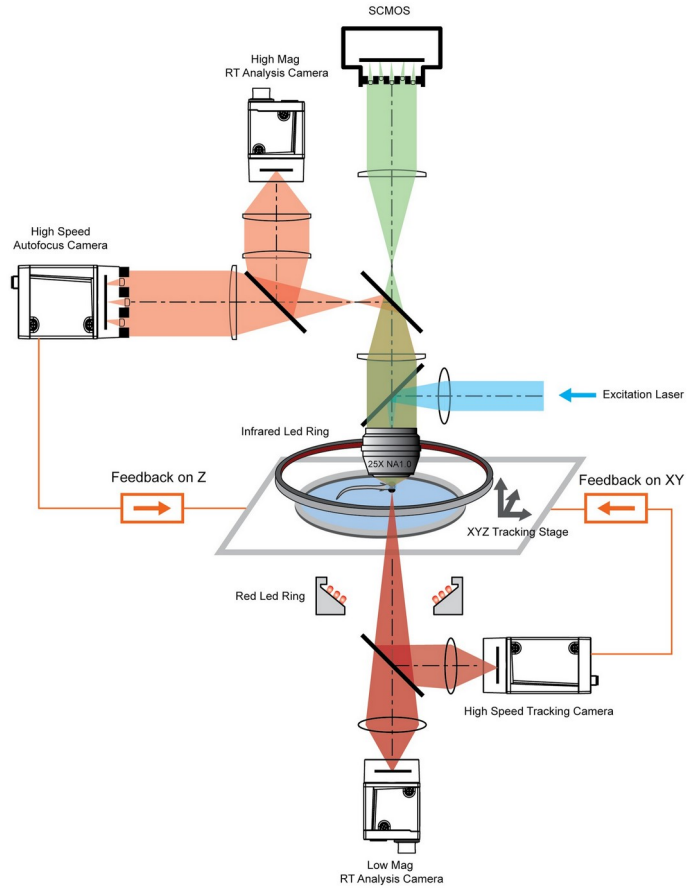
Cong et al. (2017), Video 3. Spontaneous cellular brain activity.

Tracking keeps a moving animal within the microscope

A freely moving animal changes position and orientation, which can carry its brain outside a microscope's field of view. Cong and colleagues combined brain imaging with rapid tracking of the larva. The system moved the sample stage to keep the fish's head near the imaging position during swimming.

The researchers also recorded the animal's behavior. Eye movements, body bends, and prey capture provided events against which brain activity could be compared. Recording both signals in the same animal preserves their timing, allowing a neural event to be placed before, during, or after a specific movement.

This preparation changes the anatomical question from where a signal occurs in an immobilized brain to where it occurs during action. A fish can turn its eyes toward prey before swimming toward it. Separating those events allows researchers to examine distinct stages of a sensorimotor sequence within one intact nervous system.



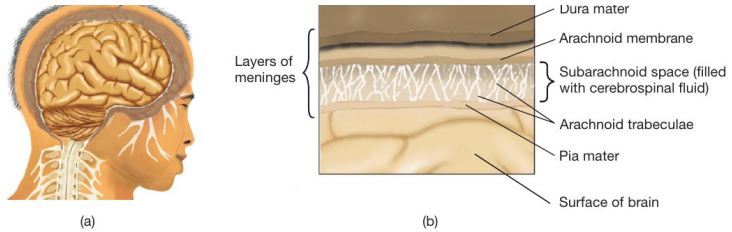
Cong et al. (2017), Fig. 2. Imaging and tracking apparatus.

Three meningeal layers surround brain and spinal cord

The **meninges** are protective membranes surrounding the brain and spinal cord. The outer **dura mater** is tough connective tissue. Beneath it lies the **arachnoid membrane**, and the delicate **pia mater** follows the surface of the nervous tissue, including the contours of its folds and grooves.

The **subarachnoid space** lies between the arachnoid and pia. It contains cerebrospinal fluid and major blood vessels that travel along the brain's surface. This fluid-filled compartment surrounds nervous tissue externally, while cavities within the brain contain another continuous portion of the cerebrospinal fluid system.

These layers define compartments that matter for injury and clinical treatment. Bleeding into one space can press on nearby tissue, and infection can inflame the membranes. An injection or surgical approach must pass through specific layers, so the location of the dura, arachnoid, and pia determines which tissues it reaches.



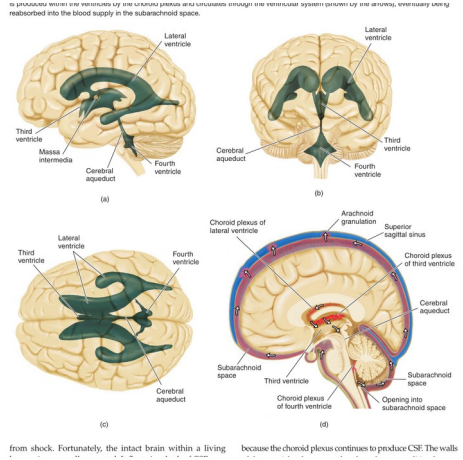
Carlson and Birkett (2022), Fig. 3.3, p. 72.

Ventricles contain cerebrospinal fluid inside the brain

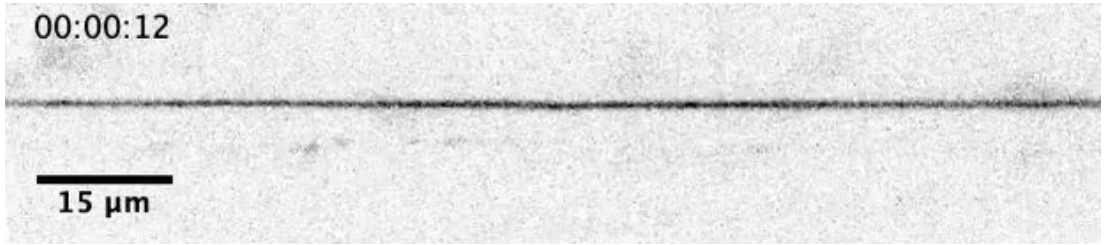
The **ventricles** are fluid-filled cavities within the brain. The two **lateral ventricles** lie within the cerebral hemispheres and communicate with the **third ventricle** near the midline. A narrow passage called the **cerebral aqueduct** carries fluid from the third ventricle toward the **fourth ventricle** beside the hindbrain.

The **choroid plexus** produces much of the cerebrospinal fluid, or **CSF**, within these cavities. Fluid passes through openings into spaces around the brain and spinal cord. Its movement involves several routes and local forces, rather than a single rigid pipe that carries all fluid at one constant speed.

The spinal cord also contains a **central canal**, continuous with the ventricular system. In larval zebrafish, a thin structure called the **Reissner fiber** lies within this canal. In the recording, the fiber moves inside a living animal, providing a direct view of a small structure within a fluid-filled central compartment.



Carlson and Birkett (2022), Fig. 3.4, p. 73.



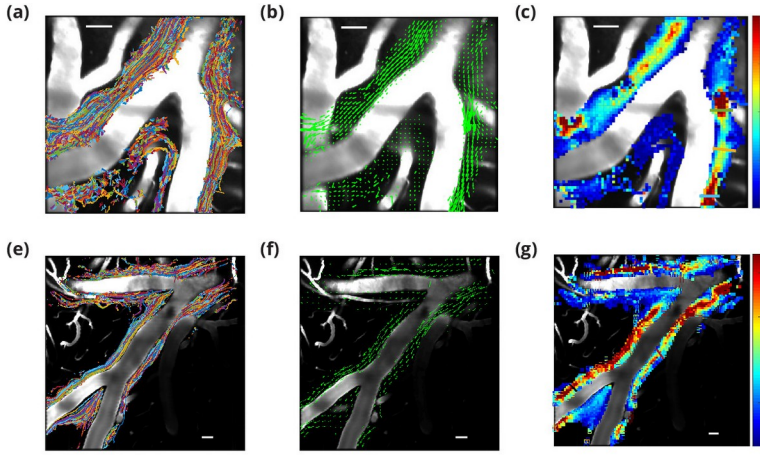
Bellegarda et al. (2023), Video 1. Reissner fiber motion.

Particles reveal fluid movement beside cortical arteries

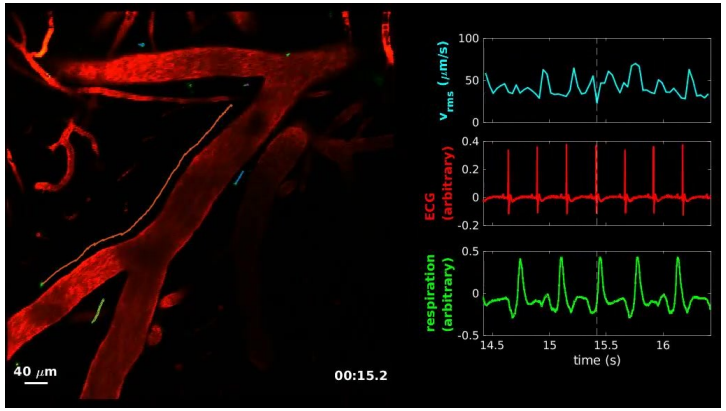
Raghunandan and colleagues followed fluorescent particles within spaces surrounding arteries on the mouse brain's surface. These **perivascular spaces** contain fluid outside the vessel wall. Tracking the particles across successive frames allowed the investigators to observe movement along the artery rather than infer movement from a single still image.

The researchers injected tracer fluid while withdrawing an equal volume through a second syringe. This approach avoided a net addition of fluid to the compartment. Particles still traveled downstream, showing that the observed transport did not require a net increase in injected volume during this experimental preparation.

The movement also varied over cardiac and respiratory cycles. Blood vessels and surrounding fluid spaces therefore participate in a mechanically active environment. The experiment demonstrates transport in the observed periarterial spaces; the particle paths identify where fluid moved and how its movement changed over time in the measured region.



Raghunandan et al. (2021), Fig. 2a–c, e–g. Perivascular particle paths and flow maps.



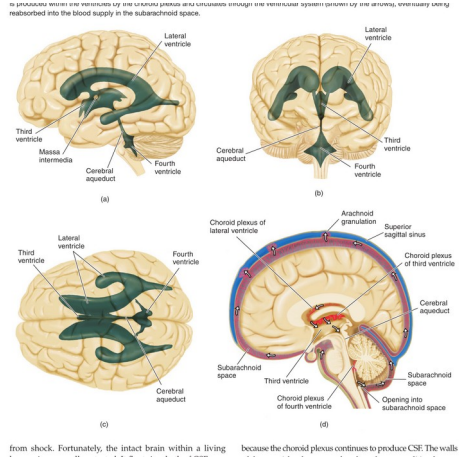
Raghunandan et al. (2021), Fig. 2, Video 1. Periarterial particle transport.

Cilia move fluid along specialized internal surfaces

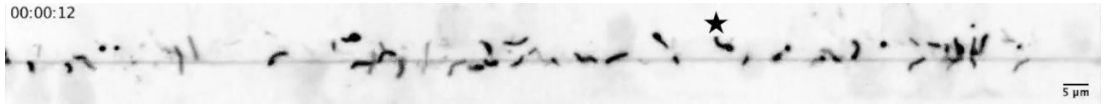
A **cilium** is a thin projection extending from a cell's surface. Some cilia beat repeatedly and exert force on surrounding fluid. Cells lining parts of the ventricular and central-canal system carry motile cilia, placing moving cellular structures directly at the boundary between nervous tissue and cerebrospinal fluid.

Ciliary movement can create local fluid motion even when the surrounding tissue remains in place. The direction and coordination of beating affect how nearby particles move. Local movement near a cilium and net transport along a larger compartment describe different spatial scales of the same fluid environment.

In the recording, cilia beat beside the Reissner fiber in a living zebrafish larva. The cilia sweep near the fiber while both remain within the central canal. This scene makes the canal's lining recognizable as an active cellular surface, rather than an empty wall around a passive reservoir of fluid.



Carlson and Birkett (2022), Fig. 3.4, p. 73.



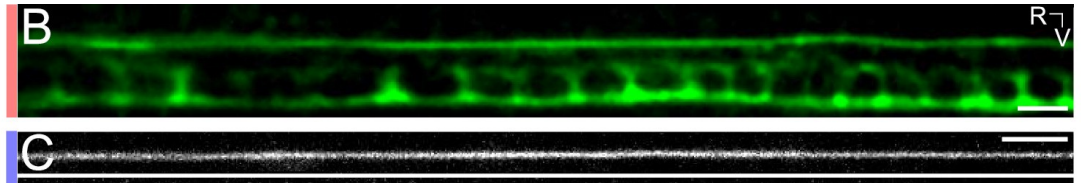
Bellegarda et al. (2023), Video 10. Ciliary motion beside the fiber.

A fiber inside the spinal canal moves in living larvae

Bellegarda and colleagues labeled the zebrafish Reissner fiber with a fluorescent protein and imaged it inside the spinal central canal. The fiber is an **acellular** thread, meaning it is not itself a neuron or a chain of cells. Its position places it beside cilia and fluid-contacting neuronal endings.

In living larvae, the fiber moved across the dorsoventral direction while remaining within the canal. The investigators compared these movements with recordings after fixation, a treatment that preserves tissue while stopping living processes. The comparison separated the dynamic behavior of the living preparation from the appearance of preserved anatomy.

The fiber also interacts with cells surrounding it. The investigators observed ciliary motion nearby and measured activity in fluid-contacting neurons. These observations established the anatomical setting for a further experiment: cutting the fiber locally and examining whether its motion and the activity of neighboring cells changed afterward.



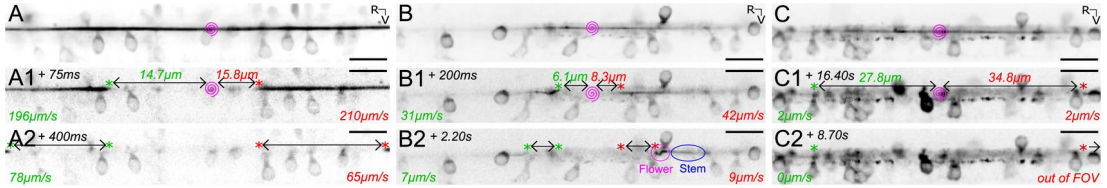
Bellegarda et al. (2023), Fig. 1B. Labeled Reissner fiber in the spinal canal.

Cutting the Reissner fiber reveals mechanical tension

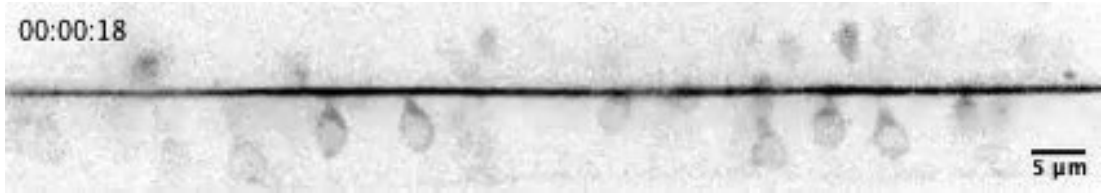
Bellegarda and colleagues used a focused laser to sever the Reissner fiber at a selected location. The cut ends retracted away from the break. A stretched elastic structure recoils when released, so the direction and speed of this movement provided evidence that the intact fiber had been under mechanical tension.

The researchers compared cuts made at different positions along the spinal axis. Retraction depended on position, with larger initial speeds after more rostral ablations. The result indicates that the fiber's mechanical state varies along its length, rather than behaving as a uniformly loose thread suspended in the canal.

Recoil begins immediately after the fiber is cut, allowing displacement to be measured across successive frames. A faster initial retraction indicates a different mechanical response from a slower one. The surrounding canal remains intact, so the fiber releases tension within the fluid and cellular environment in which it normally operates.



Bellegarda et al. (2023), Fig. 3A–C. Fiber retraction after local ablation.



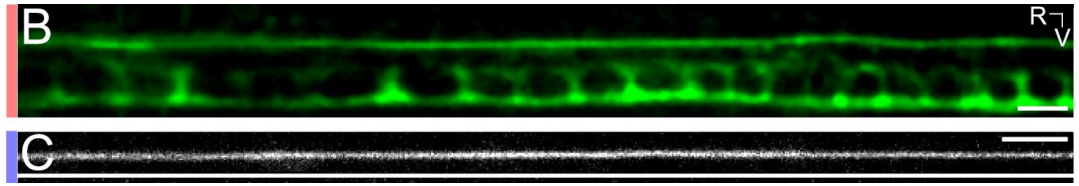
Bellegarda et al. (2023), Video 7. Fiber recoil after ablation.

Living tissue adds motion to an anatomical structure

A still image of the Reissner fiber shows where it lies, but repeated images show what it does over time. Bellegarda and colleagues recorded the same type of structure in living and fixed larvae. The moving fiber in living tissue contrasted with the largely stationary appearance after fixation.

This distinction matters because a structure's shape does not specify its mechanical behavior. A thread can look straight while bearing tension, and it can change position without being an axon that carries action potentials. The identity, location, and movement of the fiber must therefore be described as separate properties.

The investigators combined imaging with local ablation to test those properties directly. Live imaging revealed ongoing movement; cutting revealed recoil; recordings from nearby neurons measured cellular activity. Each observation answered a different question about the same anatomical compartment, building a functional account from several concrete measurements.



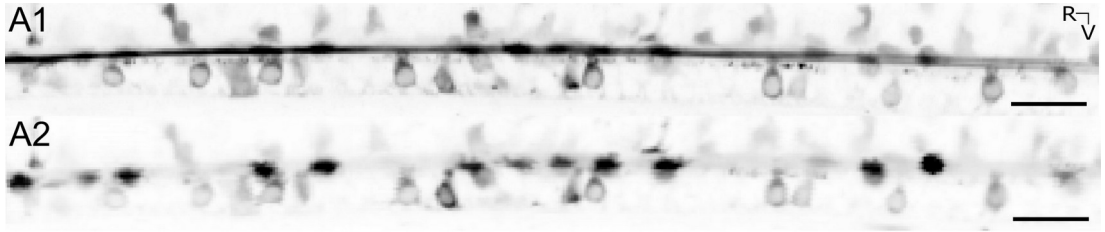
Bellegarda et al. (2023), Fig. 1B. Labeled Reissner fiber in the spinal canal.

Canal-contacting neurons sample the spinal fluid boundary

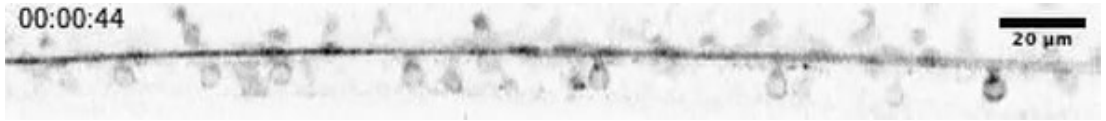
CSF-contacting neurons have a specialized ending that reaches into the central canal. Bellegarda and colleagues used a calcium indicator to record these cells in living zebrafish larvae. Their position places sensory machinery at the boundary between spinal tissue, canal fluid, and the Reissner fiber within that fluid compartment.

Before cutting the fiber, the researchers observed spontaneous calcium events in these neurons. The original recording shows the activity of labeled cells over time. A calcium event identifies a change in the recorded cellular signal; its spatial position identifies which cells responded within the surrounding spinal tissue.

The investigators then compared activity before and after a targeted fiber ablation. This sequence allowed them to test whether removing the fiber's continuity altered neuronal activity in the same anatomical system. The experiment treats the central canal as a sensory environment containing both mechanical structures and cells that detect local conditions.



Bellegarda et al. (2023), Fig. 2A1–A2. Canal-contacting neuronal calcium signals.



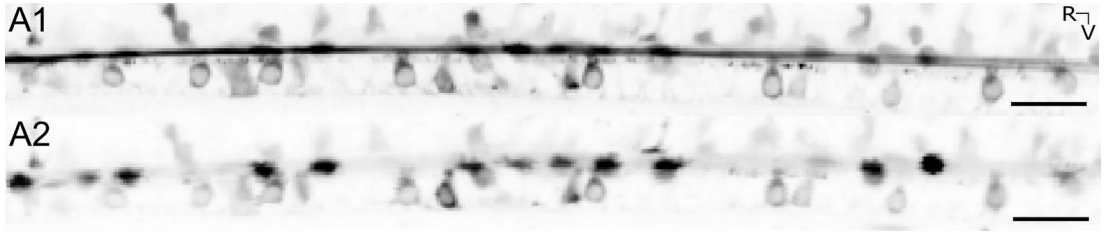
Bellegarda et al. (2023), Video 5. Neuronal activity before ablation.

Fiber ablation reduces activity in nearby sensory cells

After the Reissner fiber was cut, spontaneous calcium activity decreased in ventral CSF-contacting neurons. The researchers measured activity rather than judging the neurons only by their shape. The result shows that changing a nonneuronal structure inside the canal can alter the behavior of sensory cells bordering that compartment.

The decrease concerned spontaneous activity in cells whose endings contacted the canal. Their anatomical position places them close to the disrupted fiber, where altered mechanical interactions can change sensory input. Measuring calcium events before and after the cut tested the contribution of the intact fiber to activity in that local population.

A spinal structure can therefore contribute to signaling without being a conventional sensory nerve from the skin. The canal contains fluid, an acellular fiber, cilia, and neuronal endings. Their arrangement creates an internal sensory interface that helps explain why central anatomy includes more than collections of cell bodies and long axons.



Bellegarda et al. (2023), Fig. 2A1–A2. Canal-contacting neuronal calcium signals.



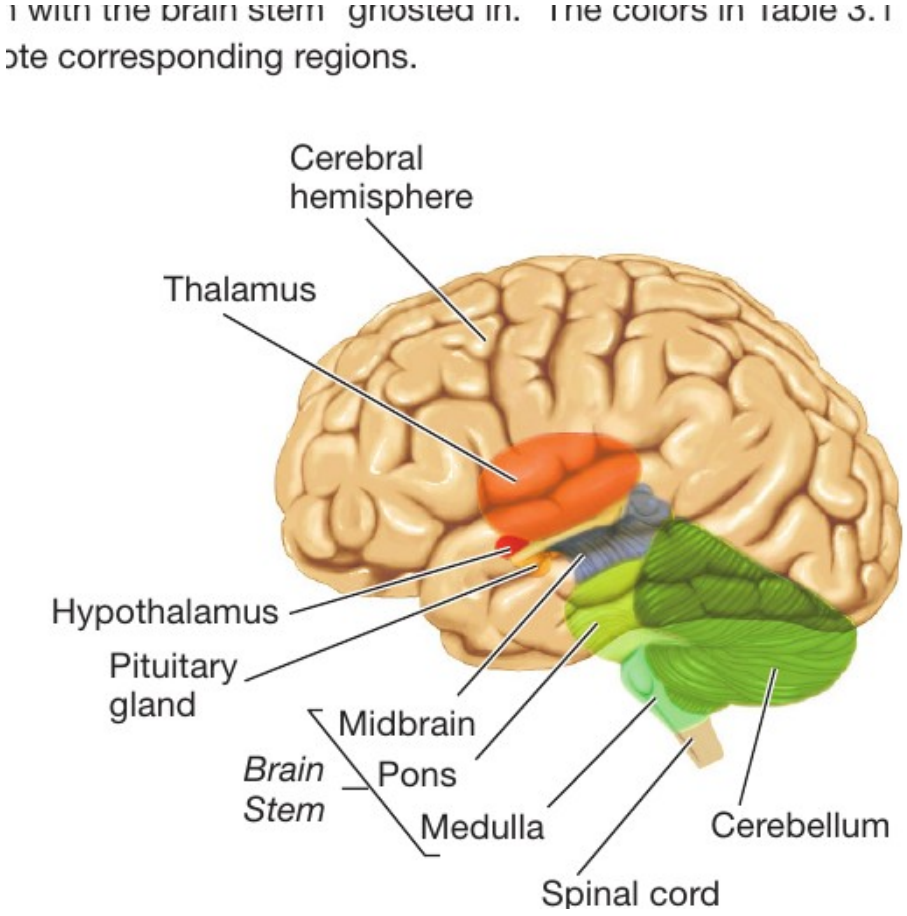
Bellegarda et al. (2023), Video 6. Neuronal activity after ablation.

Developmental divisions organize the major brain regions

The **forebrain**, **midbrain**, and **hindbrain** are major divisions established during development. In the adult brain, each division contains several recognizable structures with different cell populations and projections. Developmental names describe the origin of these regions, while their mature anatomy identifies the pathways and nuclei that operate within them.

The forebrain includes the **telencephalon** and **diencephalon**. The telencephalon contains the cerebral hemispheres, including cortex, basal ganglia, and several limbic structures. The diencephalon includes the thalamus and hypothalamus, which occupy deeper positions near the third ventricle and communicate extensively with other central regions.

The midbrain forms the **mesencephalon**. The hindbrain includes the **metencephalon**, which gives rise to the pons and cerebellum, and the **myelencephalon**, which gives rise to the medulla. These divisions organize a continuous nervous system, with pathways crossing developmental boundaries as they travel between the forebrain, brainstem, and spinal cord.



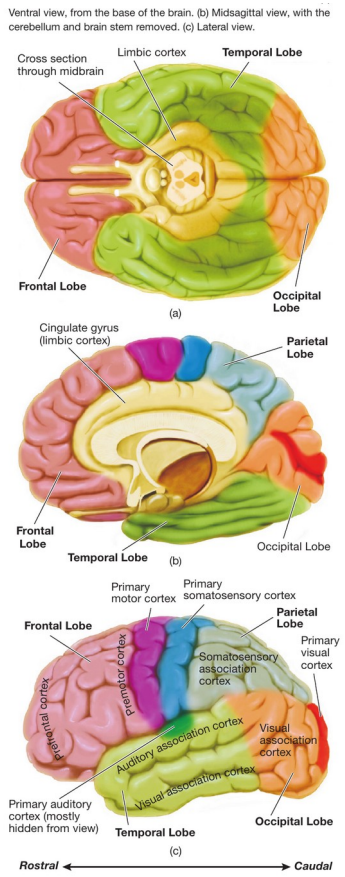
Carlson and Birkett (2022), Fig. 3.6, p. 75.

Cortical folds separate lobes without isolating their circuits

A **gyrus** is a ridge on the cerebral surface, and a **sulcus** is a groove between ridges. Deep grooves help identify the **frontal**, **parietal**, **temporal**, and **occipital lobes**. These visible landmarks provide a shared map for locating cortical areas across different views of the brain.

The frontal lobe contains primary motor cortex and regions involved in organizing behavior. Parietal regions process bodily and spatial information. Temporal regions participate in hearing, object recognition, and memory, while occipital regions contain major visual-processing areas. Each lobe contains several areas rather than serving one single psychological function.

Axons travel within and between these lobes, so a behavior usually requires communication across several regions. Recognizing an object and reaching toward it recruits visual processing, spatial information, movement planning, and motor output. The boundaries named on the surface map help locate these operations without separating them into independent organs.



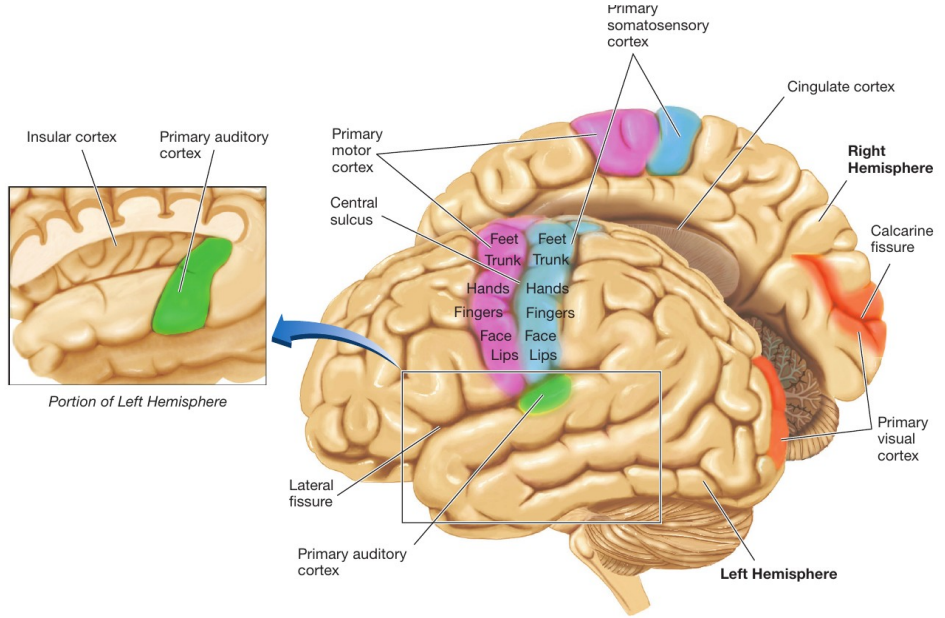
Carlson and Birkett (2022), Fig. 3.9, p. 77.

Sensory and motor areas occupy distinct cortical locations

The **primary visual cortex** lies in the occipital lobe, and the **primary auditory cortex** lies in the temporal lobe. **Primary somatosensory cortex** receives information about the body and lies just behind the central sulcus. These areas receive organized inputs that preserve important features of the sensory information.

Primary motor cortex lies just in front of the central sulcus. Its neurons contribute descending signals that influence movement through brainstem and spinal pathways. Neighboring cortical locations often influence neighboring body regions, but the map contains overlapping populations and does not function as a row of isolated muscle switches.

Association cortex processes information beyond the earliest cortical sensory stages or combines information needed for behavior. A visual target must be recognized and located before a coordinated reach can occur. Axons between cortical areas and subcortical structures allow incoming sensory information to guide movement rather than remaining a separate sensory experience.



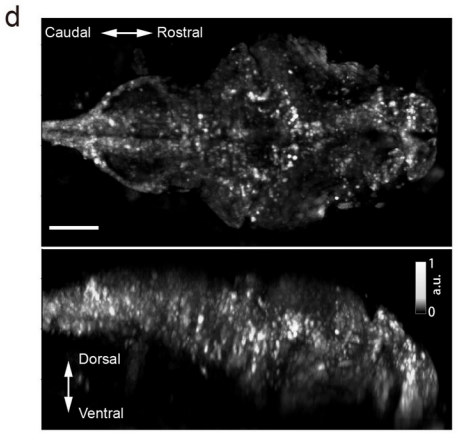
Carlson and Birkett (2022), Fig. 3.10, p. 78.

Visual stimulation recruits anatomically distributed activity

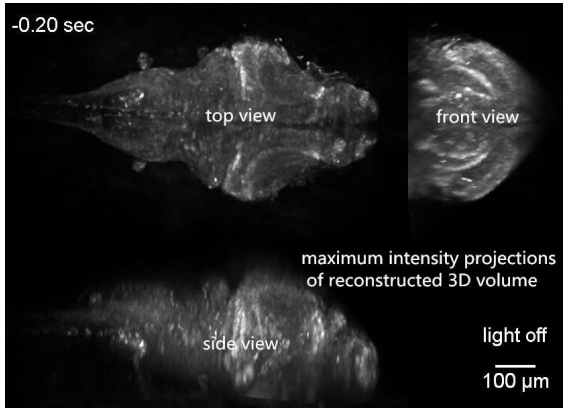
Cong and colleagues delivered light while recording calcium-dependent fluorescence across a larval zebrafish brain. The stimulus produced changes in multiple locations over time. Because the recording preserved anatomical position, the investigators could observe where activity emerged while following the temporal sequence of the response to the visual event.

The study used both nuclear and cytoplasmic indicator preparations. The accompanying recording shows a cytoplasm-labeled larva during light stimulation. Fluorescence fills a broader portion of the labeled cells than in nuclear labeling, producing a different view of the same general problem: how sensory stimulation recruits activity across an intact brain.

Sensory processing therefore extends beyond identifying one structure that receives input. Signals pass through populations that detect, transform, and act on information. Simultaneous imaging makes these distributed events observable within a single trial, placing neural changes in a shared time frame rather than assembling them from unrelated recordings.



Cong et al. (2017), Fig. 1d. Brain volume views.



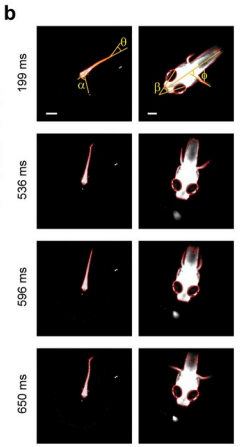
Cong et al. (2017), Video 4. Cellular activity during light stimulation.

Brain activity can be measured during free swimming

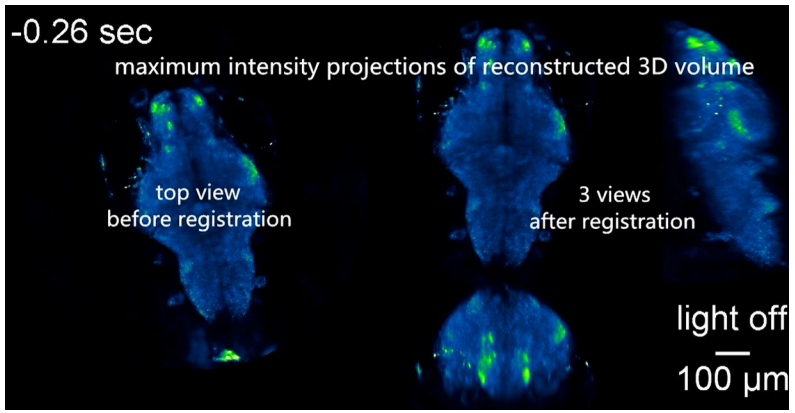
Cong and colleagues extended their imaging system to freely swimming larvae. The tracking apparatus followed the head while optical measurements recorded activity within the brain. This arrangement allowed a visual stimulus, the animal's movement, and the accompanying neural changes to be observed during the same behavioral episode.

Movement creates technical demands because the tissue changes position while the fluorescence signal also changes. The study combined rapid volume imaging with tracking to keep those changes interpretable. Anatomical registration identifies where the signal occurs; behavioral tracking identifies what the animal is doing at that moment in the recording.

During swimming, a fish changes both its location and the sensory information reaching its eyes. Neural signals can therefore reflect an external stimulus, the movement itself, or sensory consequences of that movement. Measuring activity within the behaving animal allows those events to be examined in the sequence in which they naturally occur.



Cong et al. (2017), Fig. 3b. Behavioral image sequence.



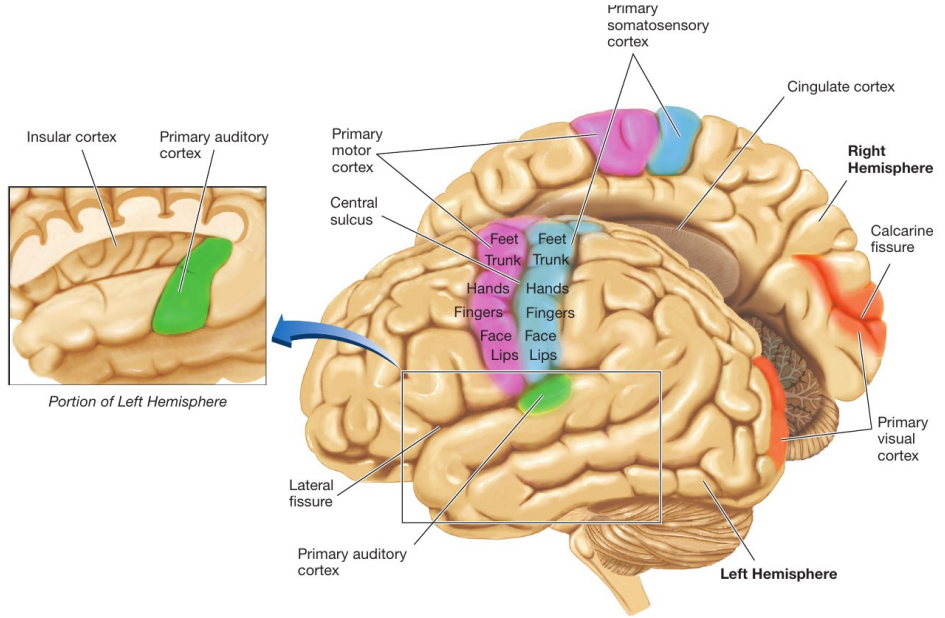
Cong et al. (2017), Video 7. Brain imaging during free swimming.

Descending pathways carry cortical signals toward muscles

A **descending pathway** carries signals from higher central regions toward lower brainstem or spinal targets. Neurons in motor cortex send axons into several such pathways. Some influence spinal circuits relatively directly, while others act through brainstem structures that organize posture, coordination, and the timing of movement.

Many motor pathways cross the midline, producing strong **contralateral** control of the body. Crossing does not mean that each hemisphere operates alone or influences only one side. Bilateral projections, brainstem pathways, and spinal interneurons allow signals from one hemisphere to affect coordinated movements involving both sides of the body.

The final command to a skeletal muscle leaves through a **motor neuron**, whose axon reaches muscle fibers outside the central nervous system. Cortical activity must therefore act through intervening circuits and motor neurons. The pathway from a cortical cell to a moving limb contains several anatomically distinct stages.



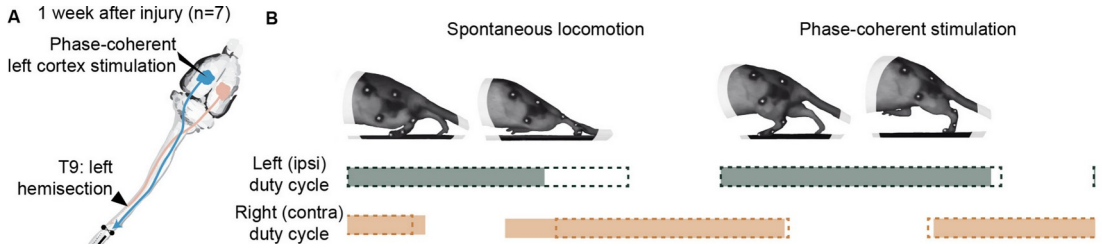
Carlson and Birkett (2022), Fig. 3.10, p. 78.

Cortical stimulation recruits the same-side hindlimb

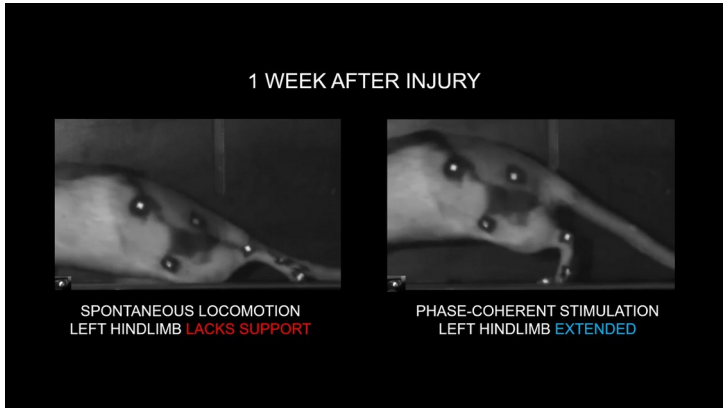
Massai and colleagues studied rats after a unilateral thoracic spinal cord injury. They stimulated motor cortex on the same side as the spinal lesion while the animals walked. The preparation tested whether surviving pathways could carry cortical influence to the impaired hindlimb through circuitry remaining after the injury.

Stimulation delivered at a selected phase of the step cycle promoted extension of the ipsilateral hindlimb along with movement of the opposite hindlimb. The response improved aspects of stepping and posture. Timing mattered because extending a limb during support serves a different mechanical purpose from extending it during swing.

The experiment demonstrates a route for same-side cortical influence under these conditions. It does not require every descending axon to remain uncrossed. Signals can pass through surviving contralateral pathways and bilateral spinal circuits, illustrating why the anatomical path between a hemisphere and a limb must be traced beyond its first projection.



Massai et al. (2024), Fig. 3A–B. Cortical stimulation during stepping.



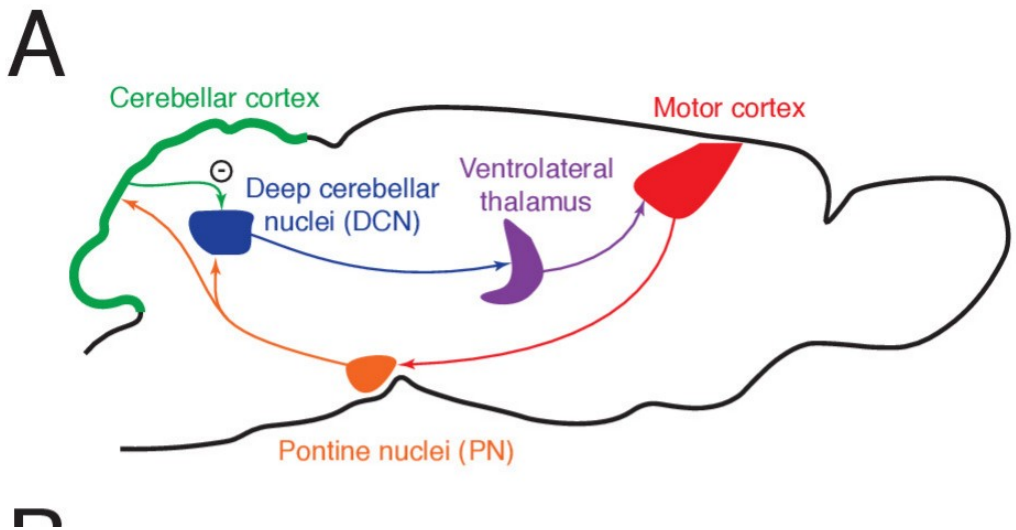
Massai et al. (2024), Fig. 3, Video 1. Stepping during cortical stimulation.

Reaching uses a loop through cortex and cerebellum

Guo and colleagues examined the pathway from motor cortex through the **pontine nuclei** to the cerebellum. These nuclei sit within the pons and relay cortical information. The cerebellum sends output onward through deep nuclei and thalamic pathways, creating a loop that can influence cortical activity during a reaching movement.

The researchers trained mice to reach toward a target and selectively stimulated the pontine nuclei. They then measured the movement and recorded activity within the loop. This approach changed a defined relay rather than removing the whole motor system, allowing the contribution of that route to be examined during behavior.

The manipulation commonly changed the precision or success of reaching while leaving movement initiation possible. The result separates starting a movement from carrying it out accurately. Anatomical relays can adjust the details of an ongoing action even when other pathways continue to generate the broad command that moves the limb.



Guo et al. (2021), Fig. 1A. Cortico-cerebellar route and relays.

Commissural axons carry signals between hemispheres

A **commissure** is a bundle of axons that crosses the midline to communicate between the two sides of the nervous system. The **corpus callosum** is the largest cerebral commissure. It contains axons traveling between cortical regions in the left and right cerebral hemispheres beneath the cortical surface.

These axons allow information processed on one side to influence activity on the other. Coordinating the hands, combining sensory information, and organizing behavior across both sides of space all require interhemispheric communication. Other commissures also cross the midline, so the corpus callosum is one major route within a broader bilateral system.

A tract image identifies the path taken by bundles of axons, while a physiological recording identifies signals carried through a preparation. These measurements answer different questions. The visible course of callosal fibers reveals where communication can travel, providing an anatomical basis for testing how activity in one hemisphere affects the other.

bundles of axons in the corpus callosum that serve different regions of the cerebral cortex.



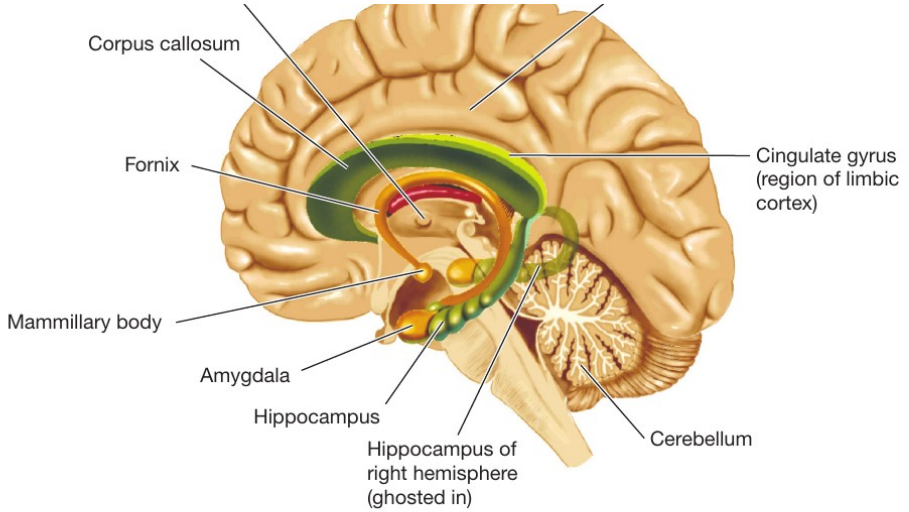
Carlson and Birkett (2022), Fig. 3.11, p. 79.

The hippocampus contributes information about context

The **hippocampus** is a curved structure within the medial temporal region. It communicates with surrounding cortex and several subcortical targets. Its neurons help represent places, events, and their context, allowing an animal to distinguish where something happened and use that information when choosing what to do later.

A **context** includes the surrounding features that make one situation different from another. The same sound or odor can have different consequences in different locations. Remembering a dangerous place therefore requires more than detecting the cue itself; it requires combining the cue with information about the environment in which it occurred.

The hippocampus does not act as a storage box separated from the rest of the brain. Its outputs influence cortical and subcortical circuits that guide action. One route reaches the hypothalamus, providing a way for information about remembered surroundings to alter defensive responses such as avoiding a location or running toward shelter.



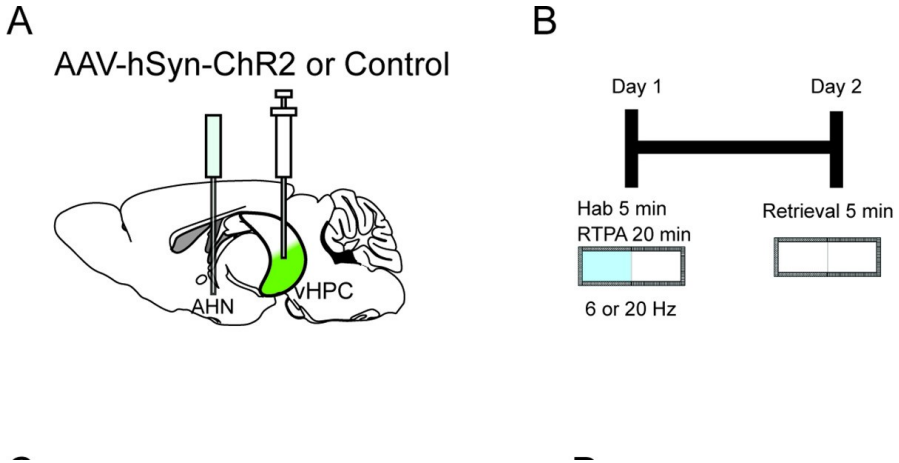
Carlson and Birkett (2022), Fig. 3.12, p. 79.

Hippocampal input helps mice avoid a remembered threat

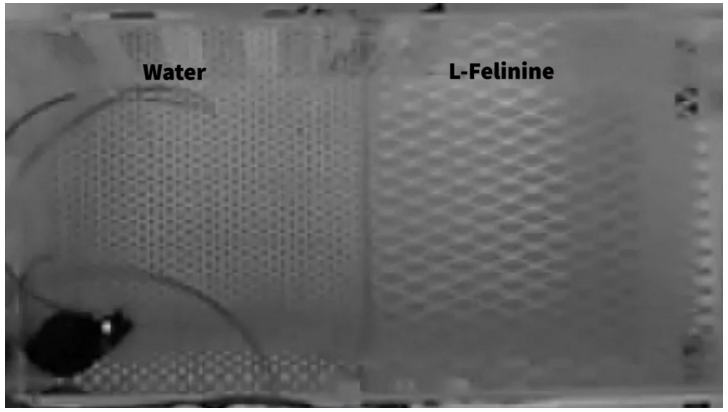
Bang and colleagues examined axons from the hippocampus to the **anterior hypothalamic nucleus**, or **AHN**. They trained mice to associate a compartment with a predator-related cue. Later behavior tested whether the animals used that contextual memory when the immediate cue no longer determined where they should move.

The researchers used **optogenetic inhibition**, in which light activates a protein that suppresses activity in selected neurons or their endings. Inhibiting the hippocampal input to the AHN impaired avoidance of the previously threatening compartment. Control animals retained stronger avoidance under the corresponding test conditions used in the experiment.

This manipulation identifies a route through which remembered context can influence defensive behavior. The hippocampus supplies information about the environment, while the hypothalamic target participates in organizing a response. Disrupting the projection changes what the mouse does in that setting, placing the anatomical pathway within a specific behavioral task.



Bang et al. (2022), Fig. 5A–B. Projection inhibition and behavioral sequence.



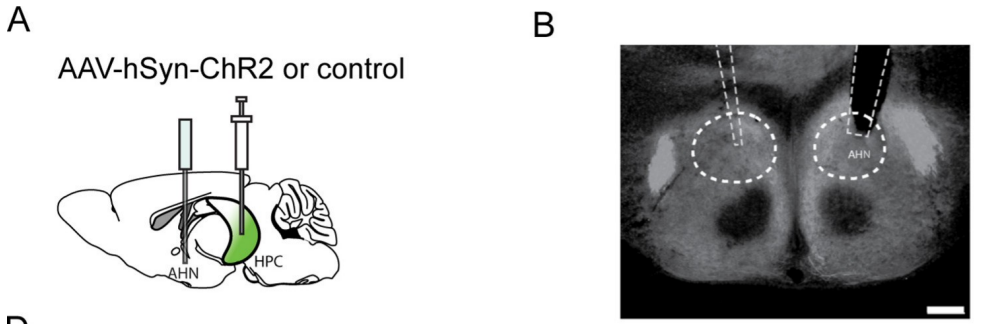
Bang et al. (2022), Video 6. Remembered threat avoidance.

Activating a defined projection changes place avoidance

Bang and colleagues also expressed **channelrhodopsin**, a light-sensitive excitatory protein, in hippocampal neurons projecting toward the AHN. Light delivered near their terminals increased activity in this selected input. The manipulation tested what the projection could produce when activated, complementing experiments that reduced its contribution during remembered threat avoidance.

Mice left a compartment in which stimulation was delivered and later avoided the location associated with that stimulation. These responses show that activating the pathway can carry a negative behavioral value. The animals' movement between compartments provides a concrete measure of how activity in the projection influences choice of location.

The pathway's source and target clarify the result. Hippocampal axons do not directly contract leg muscles; they influence hypothalamic neurons, which communicate with downstream systems controlling behavior. A change introduced at one anatomically defined input can therefore alter an entire action sequence through the circuits that receive its signals.



Bang et al. (2022), Fig. 4A–B. Targeted hippocampal axons in the AHN.



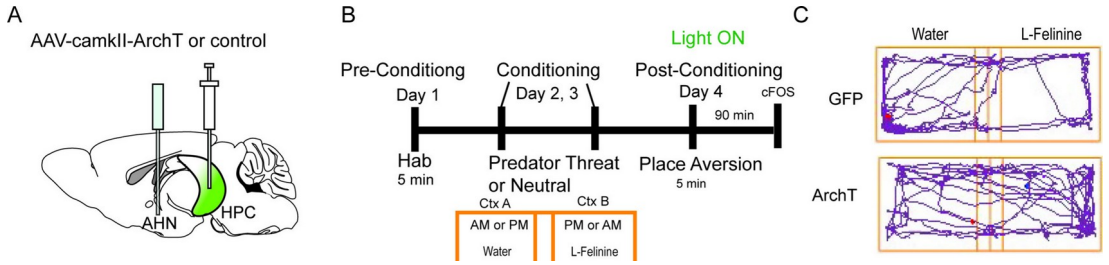
Bang et al. (2022), Video 4. Projection stimulation and place avoidance.

A shelter changes the destination of an escape response

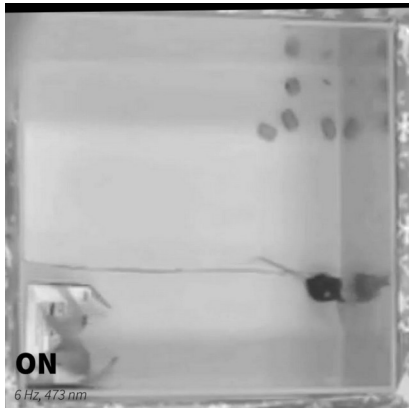
An escape response must send an animal somewhere, and available shelter can determine its destination. Bang and colleagues examined mice in an arena with a refuge. They activated the hippocampal projection to the AHN and followed the animals' paths to test how that input influenced movement toward a known safe location.

Under the study's stimulation conditions, mice moved toward the shelter. In a complementary test, inhibition of the projection impaired goal-directed escape evoked by an aversive sound. These experiments distinguish an organized route to safety from a general increase in running, because the destination and path were measured explicitly.

The result gives spatial information a role in a defensive circuit. A downstream movement system can generate locomotion, but reaching shelter also requires information about where the refuge lies. Communication between hippocampal and hypothalamic regions allows environmental knowledge to shape the direction of an action rather than only its intensity.



Bang et al. (2022), Fig. 6A–C. Shelter task and tracked escape paths.



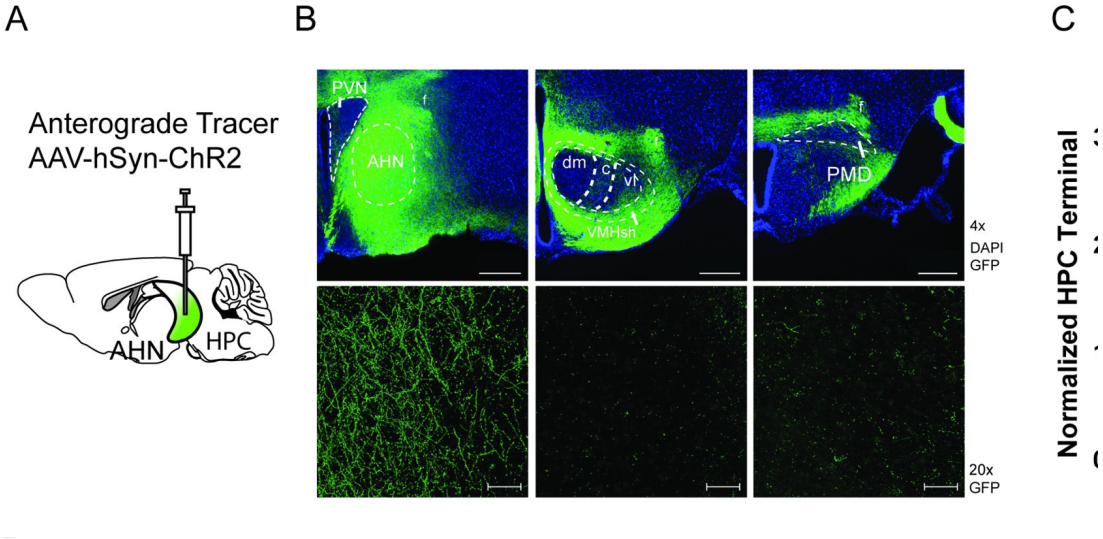
Bang et al. (2022), Video 7. Directed escape to shelter.

Tracing identifies a route from memory to defense circuits

Bang and colleagues used anatomical tracing to identify hippocampal axons entering the AHN. A **tracer** labels cells or their projections so that their locations can be examined in tissue. The labeled route placed hippocampal input within a hypothalamic system already known to participate in defensive responses in mice.

The investigators also tested transmission at this input. Activating labeled hippocampal endings evoked responses in AHN neurons, providing physiological evidence that the anatomical projection can carry signals. Seeing an axon near a target and recording a postsynaptic response supply different kinds of information about the same proposed pathway.

The study then manipulated this route during behavior. Tracing identified the source and destination, physiological recordings tested transmission, and behavioral experiments measured avoidance or escape. Together, these methods explain how a named structure contributes through a particular output rather than assigning every defensive function to the hippocampus as a whole.



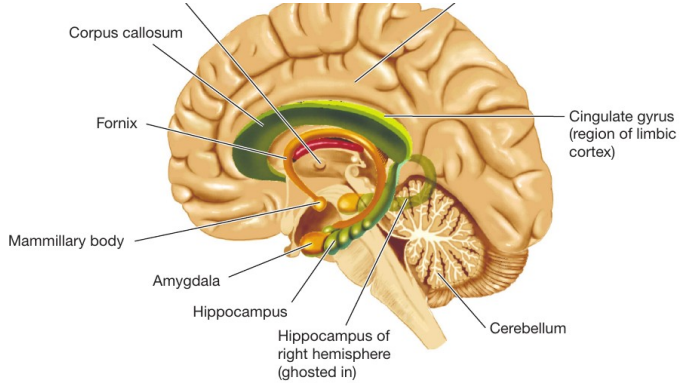
Bang et al. (2022), Fig. 3A–B. Hippocampal tracing and labeled target tissue.

Limbic structures help evaluate situations and guide action

The **amygdala** consists of several groups of neurons in the temporal region. Its circuits help evaluate biologically important stimuli and organize learning and responses to them. The amygdala communicates with cortex, hippocampus, hypothalamus, and brainstem, placing it within pathways that combine sensory information, experience, and bodily responses.

The term **limbic system** groups several interconnected structures, including the amygdala and hippocampus, but these structures do not all perform one identical function. Remembering a location, detecting a threat-related cue, and organizing an autonomic response require different computations. Their communication allows those operations to influence behavior together.

The accompanying control recording shows a mouse investigating a swab without the predator-saliva treatment used in the study. Investigation includes approach, sniffing, and movement around the object. These visible actions provide a behavioral reference for understanding how a sensory situation changes when a biologically significant chemical cue is introduced.



Carlson and Birkett (2022), Fig. 3.12, p. 79.



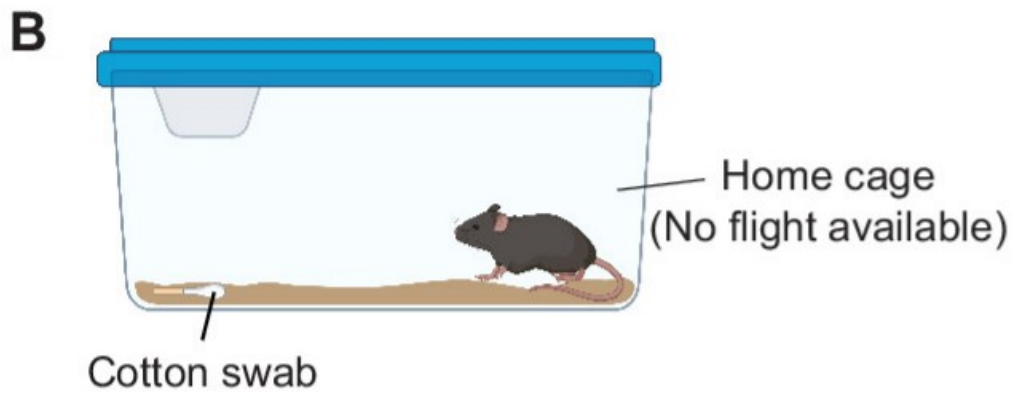
Nguyen et al. (2024), Video 1. Investigation of a control swab.

Fresh predator cues change the behavior of mice

Nguyen and colleagues exposed mice to swabs carrying cat saliva and compared their behavior with responses to control swabs. Fresh saliva produced defensive behavior, including freezing. The experiment used a specific chemical stimulus and a defined presentation procedure, allowing the animals' actions to be compared across sensory conditions.

The **vomeronasal organ** is a specialized chemical-sensing structure in the mouse nose. Its neurons send signals through an accessory olfactory pathway to forebrain regions involved in social and defensive behavior. The investigators examined activity markers along this route to identify populations recruited by predator-related chemical information.

Predator-related chemicals enter a sensory pathway before the animal changes its posture or movement. Receptor neurons in the nose detect components of the stimulus, and downstream populations process those signals. Comparing exposed tissue with control tissue identifies recruited cells along a route from the sensory surface toward central defensive systems.



Nguyen et al. (2024), Fig. 1B. Predator-cue presentation in the home cage.



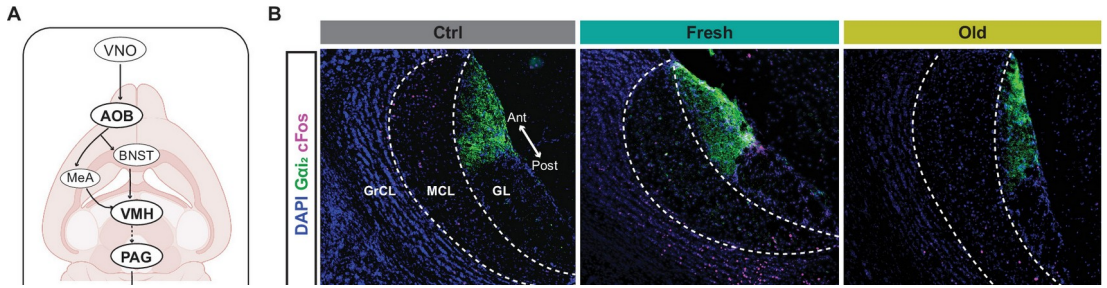
Nguyen et al. (2024), Video 2. Behavior toward fresh predator saliva.

Cue freshness changes which hypothalamic cells respond

Nguyen and colleagues compared fresh and older cat saliva. Freshness changed the defensive response and the pattern of neuronal activation. The **ventromedial hypothalamus**, or **VMH**, contained populations responding differently across the two conditions, showing that anatomically nearby cells can distinguish meaningful differences within the same general class of cue.

The researchers measured **c-Fos**, a protein often used as a marker of recent cellular activation, in tissue collected after exposure. This method locates recruited populations after an event rather than recording every spike during it. The pattern of labeled cells therefore supplies anatomical information about the response to each condition.

Fresh and older saliva recruit partly different neuronal populations in the VMH. A single named nucleus can therefore contain cells that respond differently to changes in a chemical cue. The resulting pattern provides a way for the same broad sensory route to adjust defensive behavior according to the current properties of the stimulus.



Nguyen et al. (2024), Fig. 4A–B. Sensory pathway and regional activation markers.



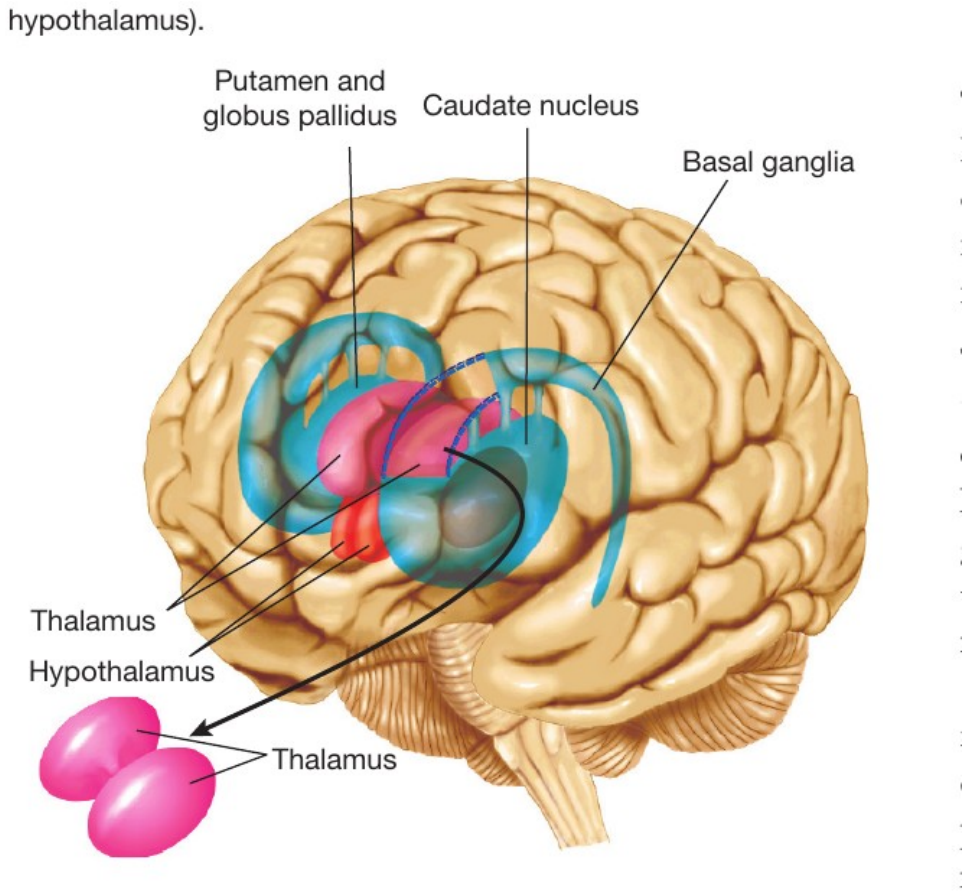
Nguyen et al. (2024), Video 3. Behavior toward older predator saliva.

Basal ganglia form deep forebrain circuits for action

The **basal ganglia** are deep forebrain structures that participate in selecting and regulating actions. Major components include the **caudate nucleus**, **putamen**, and **globus pallidus**. The caudate and putamen together form much of the **striatum**, which receives extensive input from the cerebral cortex and other regions.

These structures communicate through circuits that return toward cortex through the thalamus and also influence brainstem targets. Their activity can favor some actions while reducing competing actions. The organization helps explain why movement depends on selecting and scaling a behavior, in addition to sending a command to a particular muscle.

The **substantia nigra**, located in the midbrain, supplies dopamine to the striatum. Loss of many of these dopamine-producing neurons contributes to Parkinson's disease, including difficulty initiating and controlling movement. The affected pathway crosses a major developmental boundary, demonstrating that forebrain function depends on communication with structures outside the forebrain itself.



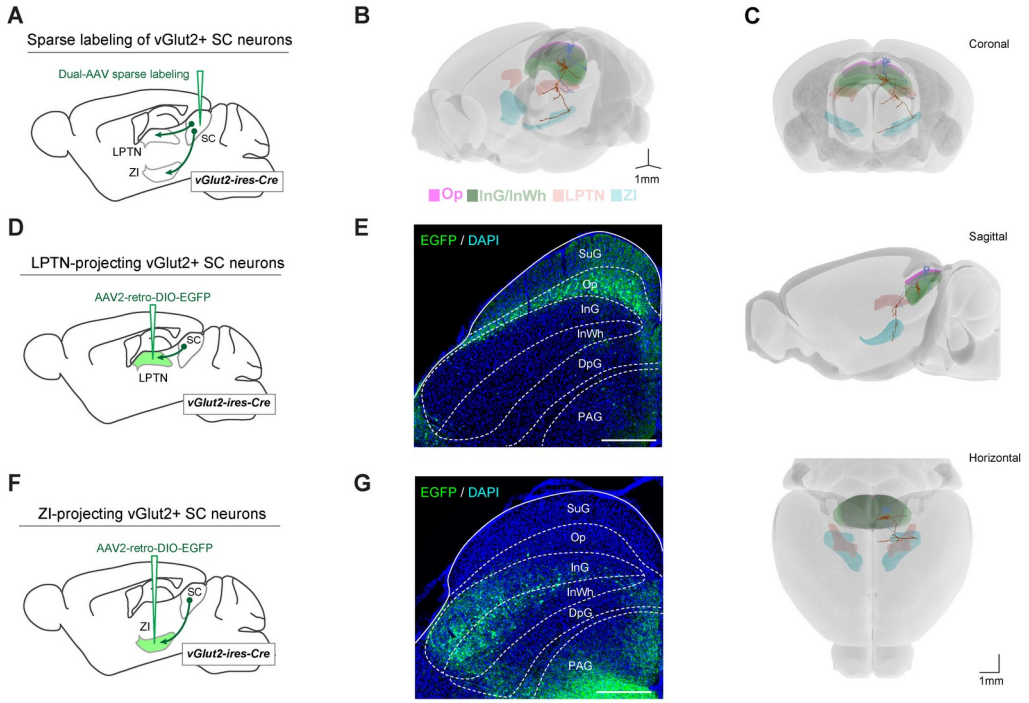
Carlson and Birkett (2022), Fig. 3.13, p. 80.

Midbrain projections reach different action-related targets

Xie and colleagues examined cell populations in the mouse **superior colliculus**, a midbrain structure involved in orienting and visually guided behavior. They combined molecular identification with anatomical tracing. Different populations sent axons to different targets, giving sensory information several possible routes toward systems that organize distinct actions.

One population identified by the gene **Pitx2** projected toward the **zona incerta**, a region beneath the thalamus. Another population identified by **Cbln2** included projections toward a thalamic target. The gene names served as markers for selecting cells, allowing projection patterns to be compared within the same larger anatomical structure.

These parallel routes show why the name of a brain region is only the beginning of a circuit explanation. Neighboring cells can send information to different destinations and contribute to different behaviors. Specifying the population and its target makes it possible to test a particular route rather than perturb every output together.



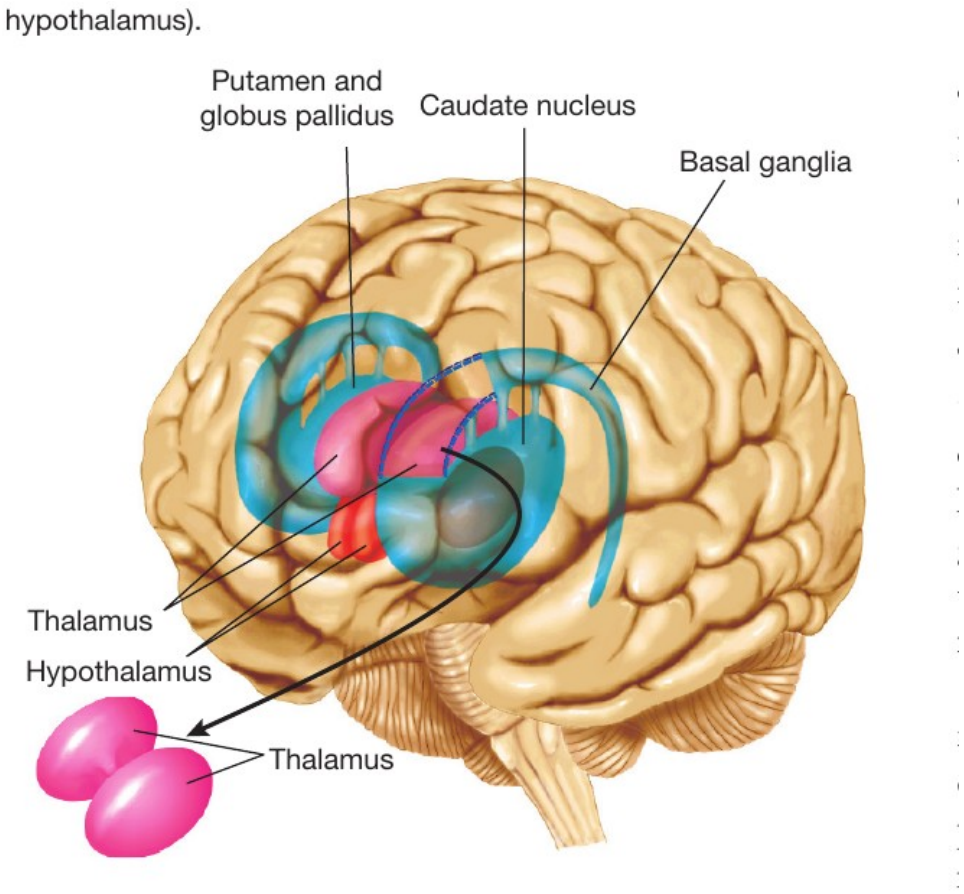
Xie et al. (2021), Fig. 2A–G. Collicular projection targets.

The thalamus routes information through specialized nuclei

The **thalamus** lies deep in the forebrain beside the third ventricle. It contains many **nuclei**, or collections of neuronal cell bodies within the central nervous system. Different nuclei receive different inputs and project to different cortical targets, creating several organized routes through the same major structure.

Visual information passes through the **lateral geniculate nucleus** toward visual cortex, while auditory information passes through the **medial geniculate nucleus** toward auditory cortex. Other nuclei carry somatosensory information or participate in motor and cognitive circuits. Most sensory systems therefore involve thalamic processing before their major cortical stages.

The thalamus also receives extensive feedback from cortex and other subcortical structures. Its neurons can alter the timing and conditions under which information reaches cortical populations. This reciprocal organization makes it an active component of distributed circuits, rather than a collection of wires that always transmits every input without modification.



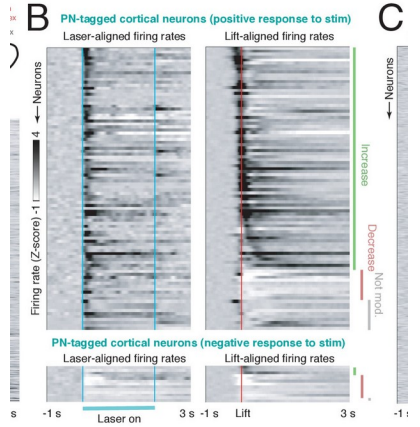
Carlson and Birkett (2022), Fig. 3.13, p. 80.

Cerebellar feedback reaches the cortex through a relay

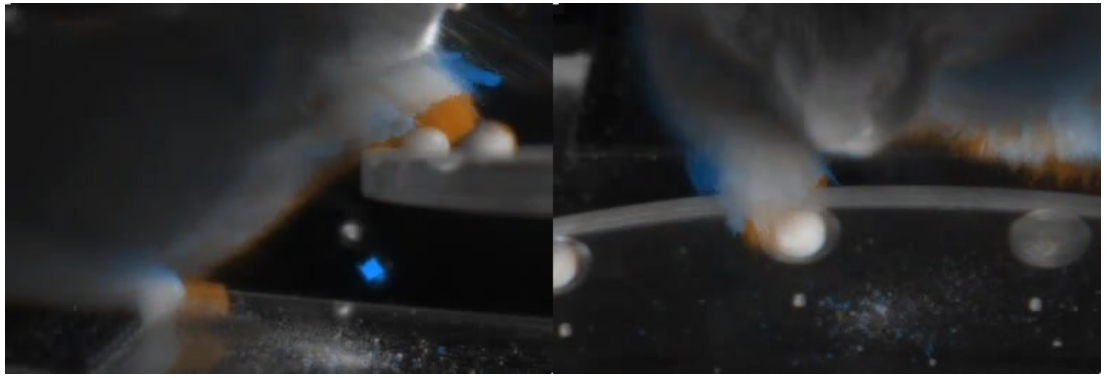
Guo and colleagues studied a motor loop in which cortical signals reach the cerebellum through the pons, while cerebellar output returns toward motor cortex through the thalamus. They identified cortical neurons receiving feedback from this system and examined their activity during a trained reach-to-grab behavior in mice.

Stimulation of the pontine relay altered movement-related activity and the details of the reach. Much of the broader neural response remained present, but differences in activity carried information about differences in hand velocity. The loop therefore contributed to movement execution while other descending routes continued to influence the limb.

Perturbing the pontine relay changes the input arriving at the cerebellum, which can then change feedback reaching cortex through the thalamus. The resulting hand movement depends on activity around this loop and on descending outputs elsewhere. The repeated reaches place a specific anatomical relay within a measurable problem of movement accuracy.



Guo et al. (2021), Fig. 4B. Cortical neurons receiving pontine-cerebellar feedback.



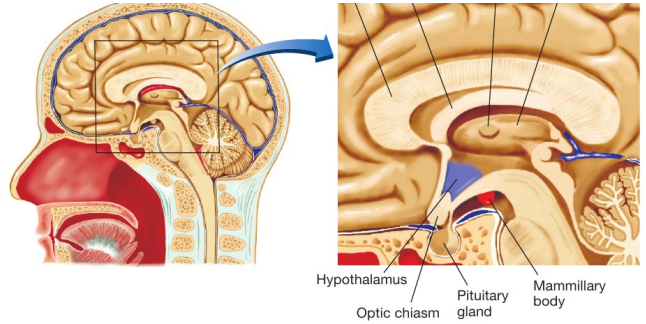
Guo et al. (2021), Video 1. Recorded reaches during pontine stimulation.

The hypothalamus coordinates behavior and internal state

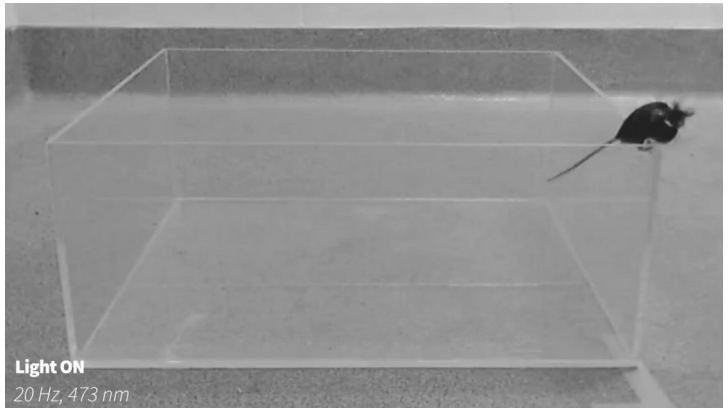
The **hypothalamus** lies below the thalamus around the lower portion of the third ventricle. Its nuclei regulate feeding, drinking, temperature, reproduction, and responses to danger. These functions require coordination between behavior and the body's internal conditions, rather than control of a single sensory modality or one group of muscles.

Hypothalamic outputs influence the **autonomic nervous system**, which regulates organs, and the **endocrine system**, which releases hormones into the blood. Other projections reach brainstem circuits organizing movement and defensive responses. Different outputs allow a change in environmental conditions to recruit bodily adjustments and overt actions together.

The anterior hypothalamic nucleus participates in defensive behavior, but neighboring hypothalamic populations have different targets and functions. Activating a selected population can alter locomotion or posture through descending pathways. The direction of its axons determines which downstream circuits receive that signal and which components of the bodily response it can influence.



Carlson and Birkett (2022), Fig. 3.14, p. 80.



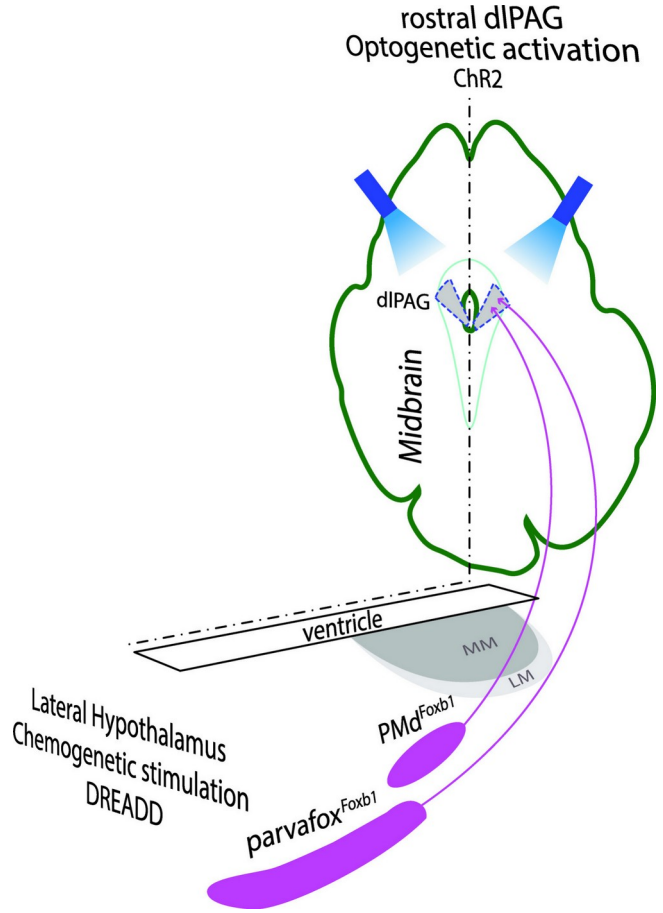
Bang et al. (2022), Video 1. Behavior during AHN stimulation.

Hypothalamic neurons project to a midbrain defense region

Cola and colleagues examined hypothalamic neurons expressing **Foxb1**, a gene used to identify a selected population. These neurons included cells in the parvafox region and dorsal premammillary nucleus. Their descending axons reached the **dorsolateral periaqueductal gray**, a midbrain region positioned around the cerebral aqueduct.

The investigators used anatomical labeling to identify the cells and their projections, then manipulated the selected population or its terminals. This approach separated the source of the pathway from its downstream target. A projection can be tested at its ending, where released neurotransmitter acts on cells in the receiving region.

The preparation allowed behavior, breathing, and heart rate to be measured after activation. These outputs involve different organs and motor systems, so observing them together tests whether a central pathway can coordinate several components of a bodily response. The anatomical route provided the specific entry point for that experiment.



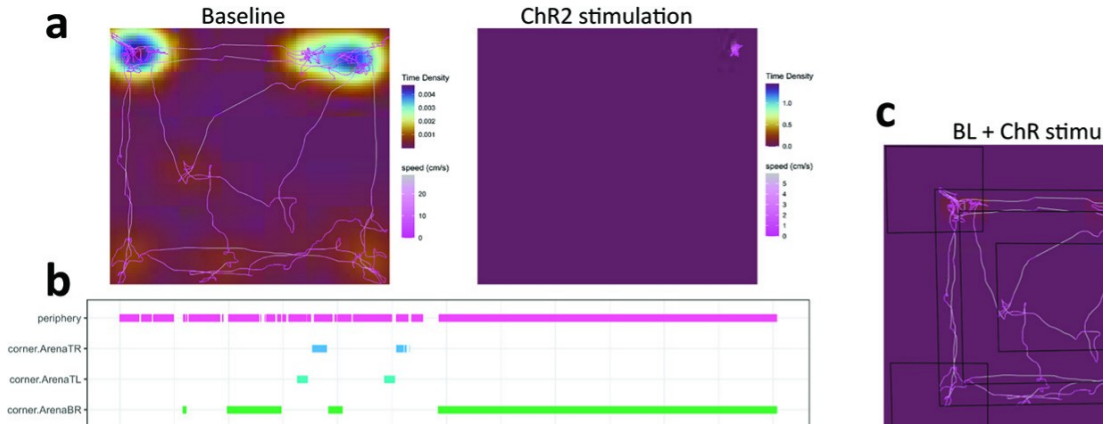
Cola et al. (2024), Fig. 1. Hypothalamic projections to the dorsolateral PAG.

Hypothalamic terminal activation changes defensive posture

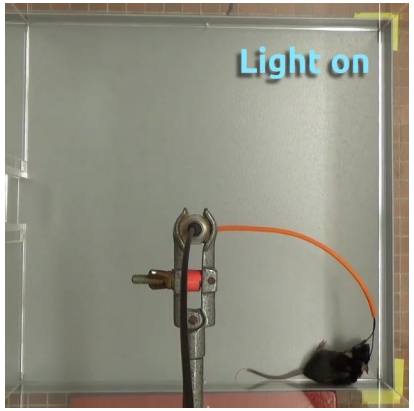
Cola and colleagues activated Foxb1-expressing hypothalamic axon terminals in the dorsolateral periaqueductal gray. The mice became largely immobile during stimulation, although some continued to move their heads. The manipulation therefore altered the organization of movement rather than simply preventing every muscle in the body from producing any movement.

The study also measured faster breathing and slower heart rate under selected activation conditions. These responses show that one central pathway can recruit distinct bodily adjustments at the same time. Respiratory movements, cardiac regulation, and limb behavior depend on different downstream components even when a shared upstream signal helps coordinate them.

Immobility and ongoing head movement can occur together because a defensive posture recruits selected motor patterns. The limb muscles need not be paralyzed for locomotion to stop. At the same time, descending activity can adjust respiratory and autonomic circuits, producing internal changes that require physiological measurements alongside observation of posture.



Cola et al. (2024), Fig. 3a–b. Movement during terminal stimulation.



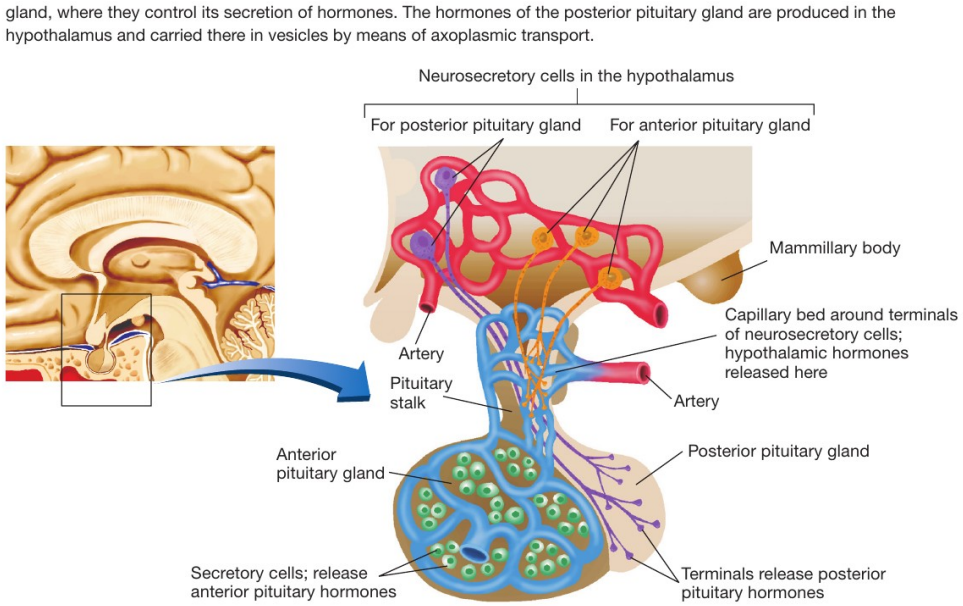
Cola et al. (2024), Video 1. Posture during pathway stimulation.

The pituitary provides two routes from brain to hormones

The **pituitary gland** hangs beneath the hypothalamus. Its **posterior** portion contains terminals of hypothalamic neurons whose cell bodies lie in the brain. These terminals release hormones into the bloodstream, allowing neuronal activity to influence distant organs through a chemical signal carried outside the nervous system.

The **anterior pituitary** contains endocrine cells controlled by releasing and inhibiting hormones from the hypothalamus. These signals enter a specialized blood-vessel route before reaching the gland. Anterior pituitary cells then release their own hormones, some of which regulate other endocrine glands elsewhere in the body.

These two routes differ in their cellular arrangement. Posterior pituitary release comes from hypothalamic neuronal endings, while anterior pituitary release comes from endocrine cells receiving hypothalamic hormonal signals. Both connect central regulation with the bloodstream, allowing the nervous system to change bodily conditions on a broader spatial and temporal scale.



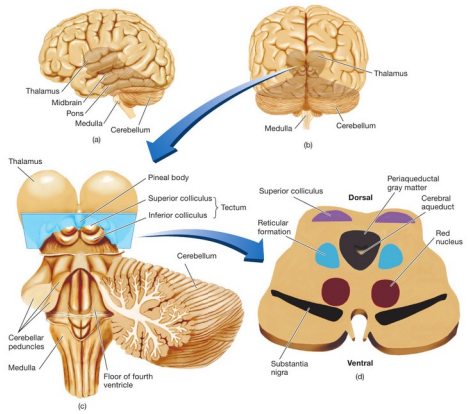
Carlson and Birkett (2022), Fig. 3.15, p. 81.

The midbrain contains sensory and motor regions

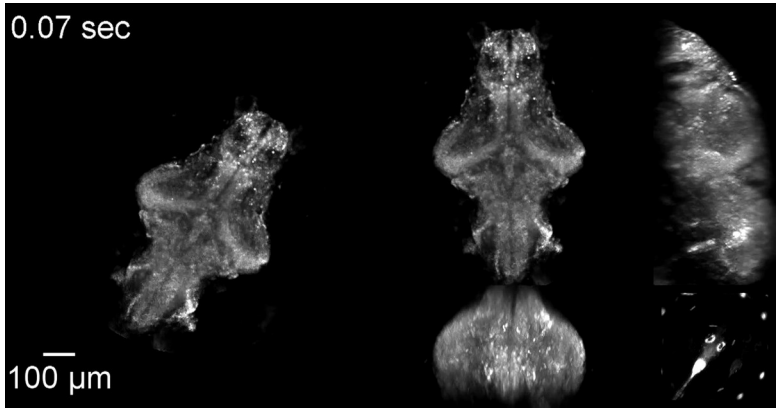
The **midbrain** surrounds the cerebral aqueduct between the forebrain and hindbrain. Its dorsal region, the **tectum**, contains the superior and inferior colliculi. The **superior colliculus** participates in orienting to sensory targets, while the **inferior colliculus** is an important stage in pathways processing auditory information.

The **tegmentum** lies below the tectum and contains several nuclei and pathways involved in movement, arousal, and other functions. The periaqueductal gray surrounds the aqueduct, while the substantia nigra lies farther ventrally. These landmarks locate structures that have different targets despite occupying the same broad brain division.

The accompanying recording shows a larval zebrafish capturing prey while its brain is imaged. Fish use a related tectal visual system to guide orientation toward targets. The scene makes the sensory-to-action problem concrete: the nervous system must locate a small moving object and direct eyes and body toward it.



Carlson and Birkett (2022), Fig. 3.16, p. 82.



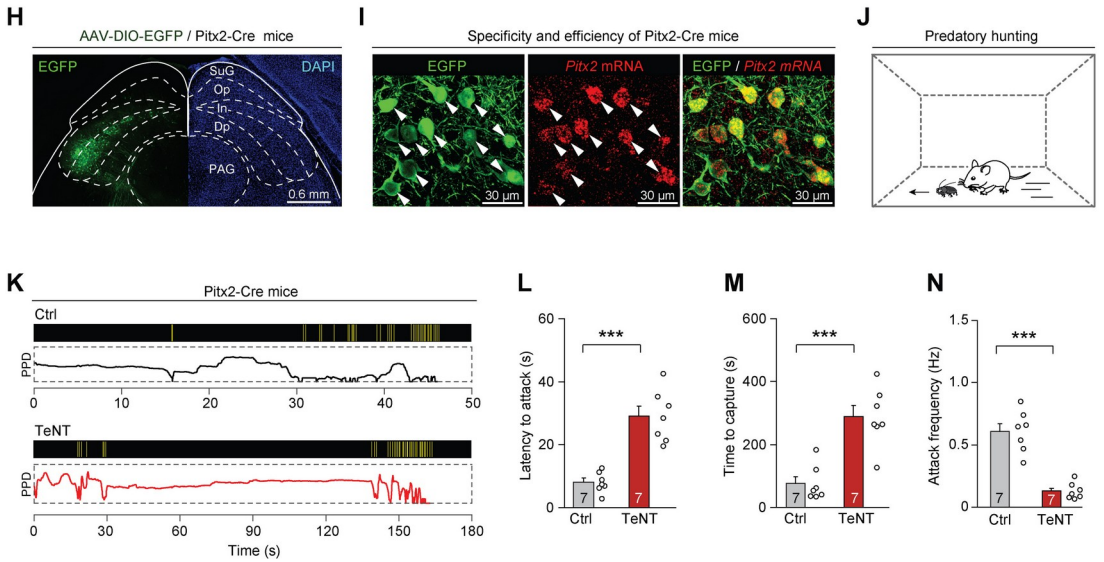
Cong et al. (2017), Video 9. Prey capture in a nuclear-labeled larva.

Selected collicular cells contribute to prey capture

Xie and colleagues identified superior-colliculus populations with different molecular markers and projection patterns. They selectively blocked synaptic output from Pitx2-expressing cells using **tetanus toxin light chain**, a protein that disrupts transmitter release. This manipulation tested the contribution of one population while preserving many neighboring cells and pathways.

Mice with this selected output blocked showed impaired prey capture in the study's arena. The experiment therefore assigned a behavioral contribution to a defined subset of collicular neurons. The relevant unit was a cell population within the structure, rather than the entire superior colliculus treated as one uniform relay.

A different population, identified by Cbln2, contributed to visually evoked freezing. The anatomical and behavioral separation shows how a sensory structure can route information toward different actions. Detecting an object does not uniquely specify what to do; the responding population and its output pathway help determine the resulting behavior.



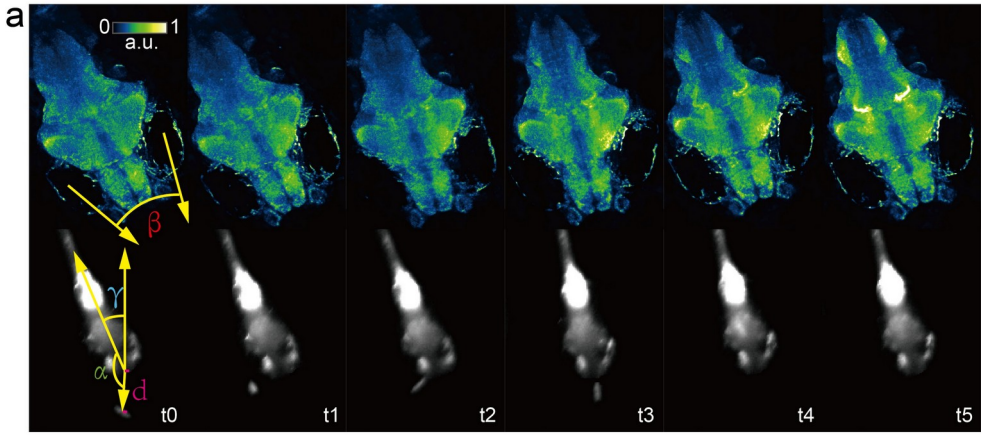
Xie et al. (2021), Fig. 3H–P. Pitx2-positive cells and prey-capture measurements.

Prey capture combines eye orientation and swimming

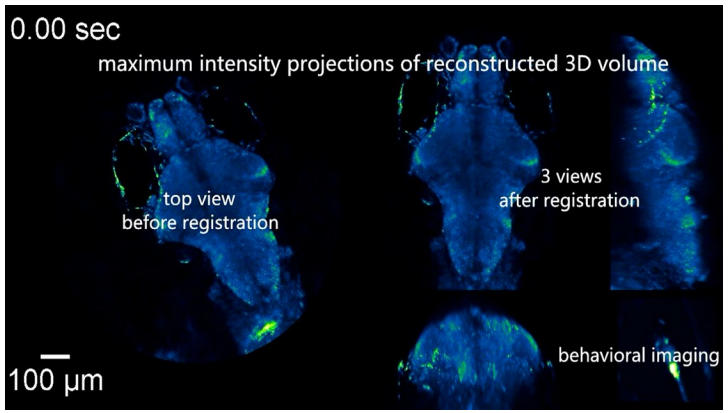
Cong and colleagues imaged larval zebrafish during actual prey capture. The animal's eyes converged as it approached a small prey item, and swimming brought the body into position. These observable events divide one successful action into stages that can be aligned with neural activity recorded in the same animal.

A target's position changes as the fish moves. Sensory information must therefore be updated while motor commands are being executed, rather than calculated once before movement begins. Recording neural activity together with the evolving behavior makes it possible to examine signals during this changing sensory and mechanical situation.

The fish brings its eyes toward the prey, changes body orientation, and then produces a capture movement. Those stages place different demands on sensory and motor circuits. A useful account of prey capture must therefore identify activity associated with each event, rather than treating the entire sequence as one undifferentiated burst of movement.



Cong et al. (2017), Fig. 4a. Brain activity and prey-capture sequence.



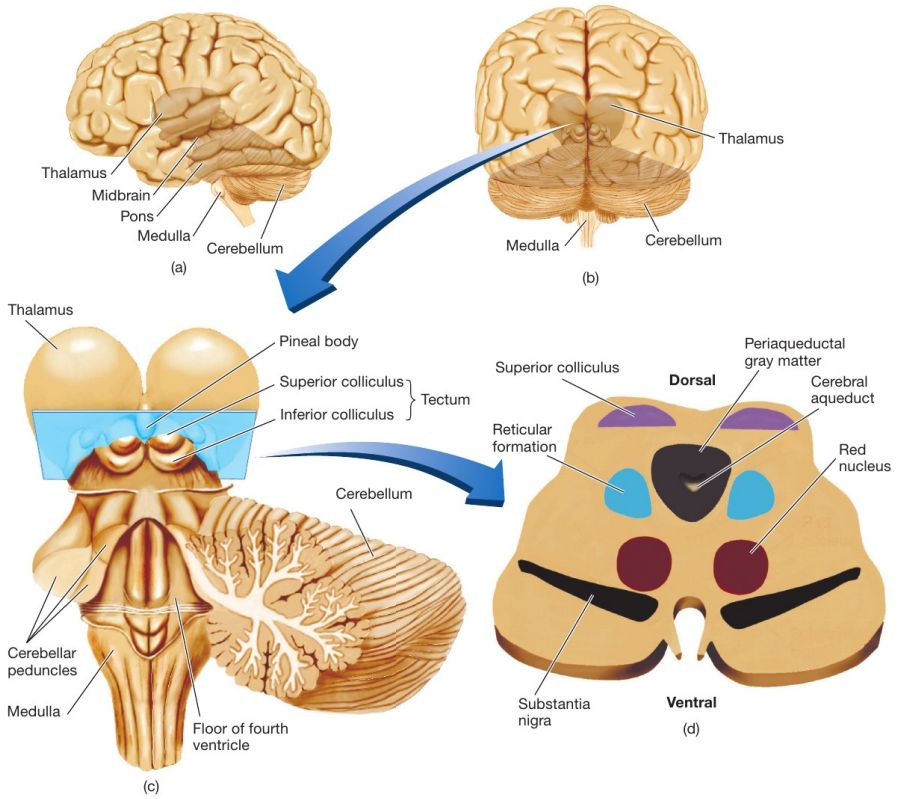
Cong et al. (2017), Video 8. Prey capture in a cytoplasm-labeled larva.

The periaqueductal gray coordinates protective responses

The **periaqueductal gray**, or **PAG**, is gray matter surrounding the cerebral aqueduct in the midbrain. Different portions participate in defensive behavior, pain modulation, and autonomic adjustments. Its position beside the aqueduct identifies its location; the axons entering and leaving it explain how it influences other systems.

Inputs reach the PAG from hypothalamic, cortical, and other subcortical regions. Outputs influence brainstem systems that regulate movement, breathing, and pain transmission. This arrangement allows information about threat or bodily state to recruit coordinated responses through downstream circuits rather than requiring the PAG to contact every muscle or organ directly.

Pain modulation means changing the transmission or processing of pain-related information. PAG activity can recruit descending pathways that alter signaling in lower regions, including spinal circuits. A stimulus reaching the nervous system can therefore produce different experiences and actions depending on activity in systems that regulate its later transmission.



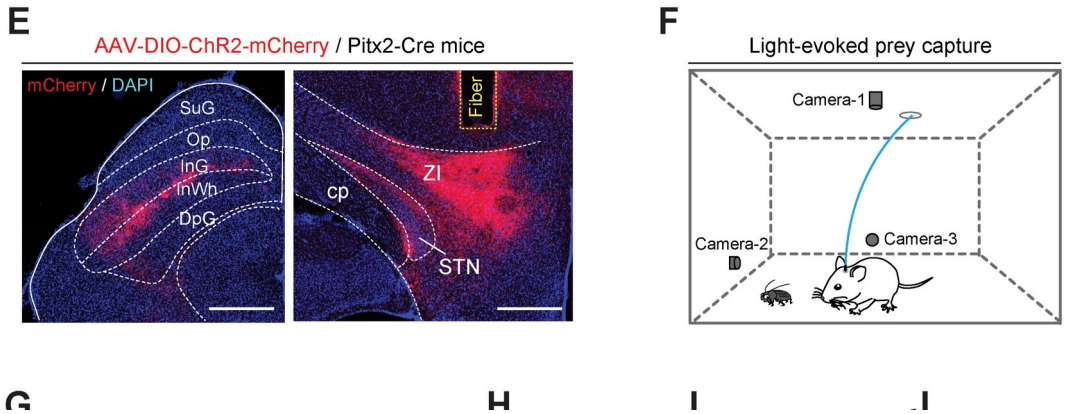
Carlson and Birkett (2022), Fig. 3.16, p. 82.

Distinct midbrain outputs favor different defensive actions

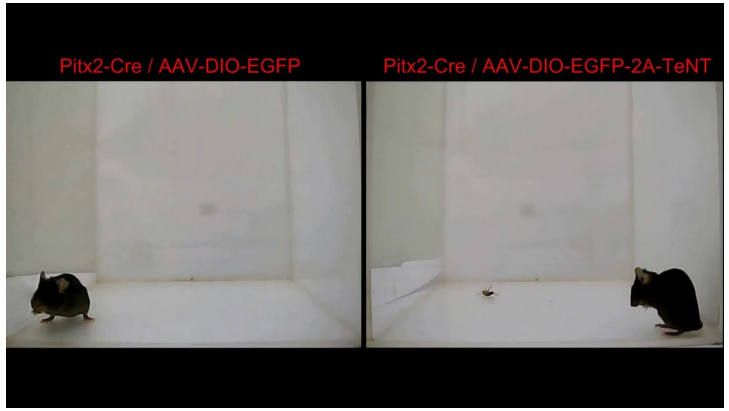
Xie and colleagues tested specific superior-colliculus projections rather than activating every collicular neuron together. Stimulation of Cbln2-positive terminals in a thalamic target induced freezing, while stimulation of Pitx2-positive terminals in the zona incerta promoted hunting behavior. Different output routes therefore carried different influences on action within the same sensory system.

The accompanying original recording shows impaired prey capture after synaptic output from Pitx2-positive collicular neurons was blocked. That loss-of-function condition complements the projection-stimulation experiments. One manipulation tests what activation can produce; the other tests how behavior changes when a selected population can no longer release transmitter normally.

The distinction between source cells and projection target is essential for explaining the result. Two populations can receive information from the same sensory environment while sending it to different downstream circuits. Naming both ends of the route explains how a common input situation can be transformed into different organized behaviors.



Xie et al. (2021), Fig. 8E–F. Collicular axons in zona incerta and hunting task.



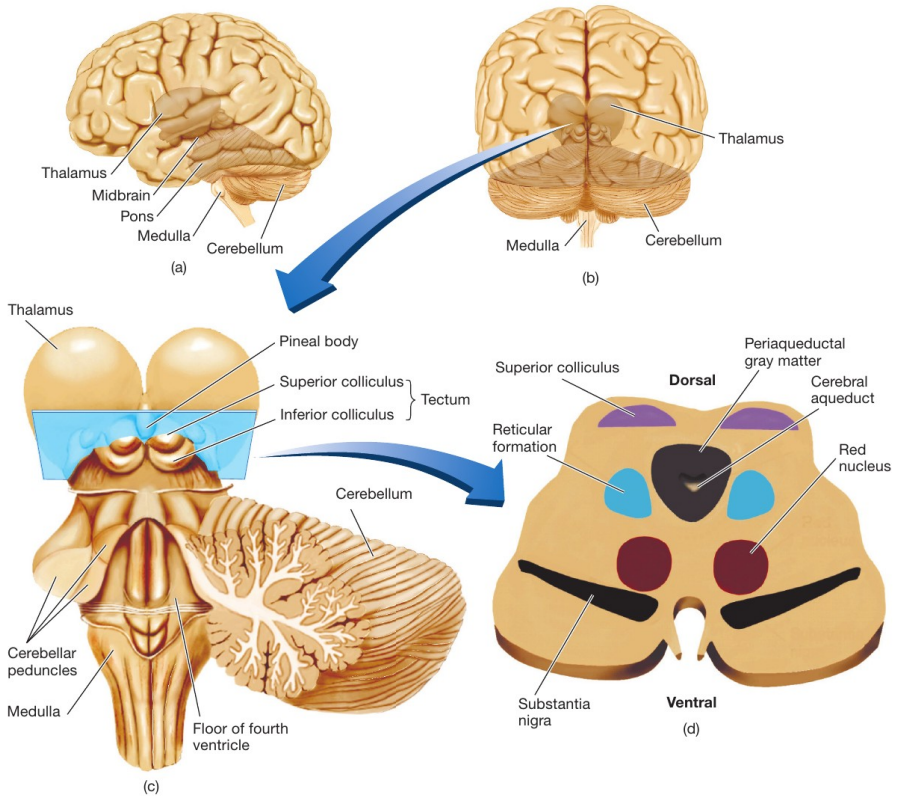
Xie et al. (2021), Fig. 3, Video 2. Prey capture after selected synaptic blockade.

The hindbrain contains the pons medulla and cerebellum

The **hindbrain** contains the pons, medulla, and cerebellum. The pons and medulla form part of the **brainstem**, the continuous central region between the forebrain and spinal cord. The cerebellum lies behind the brainstem and communicates with it through large bundles of axons called cerebellar peduncles.

The **pons** contains relays carrying cortical information toward the cerebellum, along with nuclei involved in other functions such as sleep and cranial-nerve activity. The **medulla** contains circuits regulating breathing, cardiovascular function, and protective reflexes. Long ascending and descending pathways also pass through these regions between more distant targets.

The **reticular formation** extends through much of the brainstem rather than forming one isolated round nucleus. Its populations contribute to arousal, posture, and motor regulation. A brainstem injury can therefore affect several functions at once by damaging local nuclei and pathways passing through the same compact anatomical region.



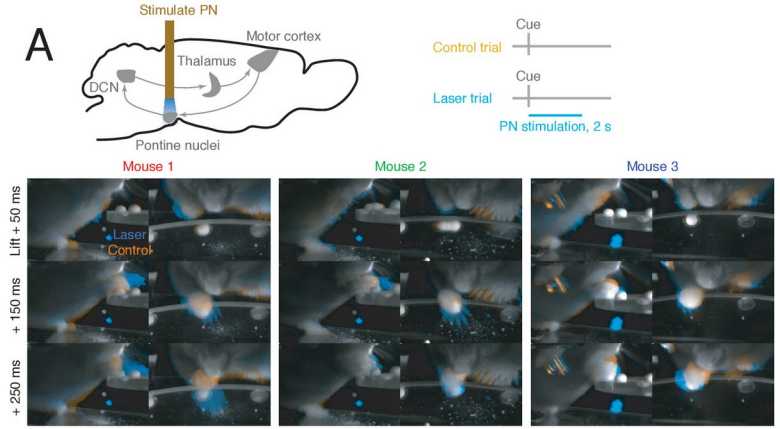
Carlson and Birkett (2022), Fig. 3.16, p. 82.

Changing a pontine relay disrupts skilled reaching

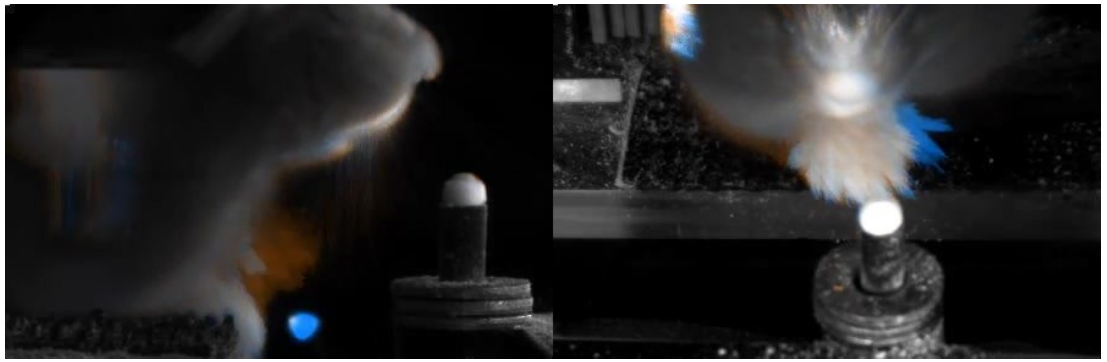
Guo and colleagues stimulated pontine nuclei while mice performed a trained reaching task. The intervention altered the signals entering the cerebellar pathway from cortex. In many trials the animal still initiated a reach, but the resulting movement became less accurate, less precise, or less likely to succeed at the target.

In one examined animal, pontine stimulation strongly interfered with reach initiation, while other animals still reached with altered movement details. The same relay can therefore affect several components of the action under different conditions. Initiation, trajectory, and successful contact with the target are distinct behavioral measurements within the trained task.

The result assigns a concrete role to a relay in the pons. Cortical output reaches several targets, so perturbing one branch can change details of behavior without eliminating every movement command. The cortico-cerebellar route contributes to executing a dexterous action through its effect on the timing and pattern of activity.



Guo et al. (2021), Fig. 5A. Original reaching frames and perturbation task.



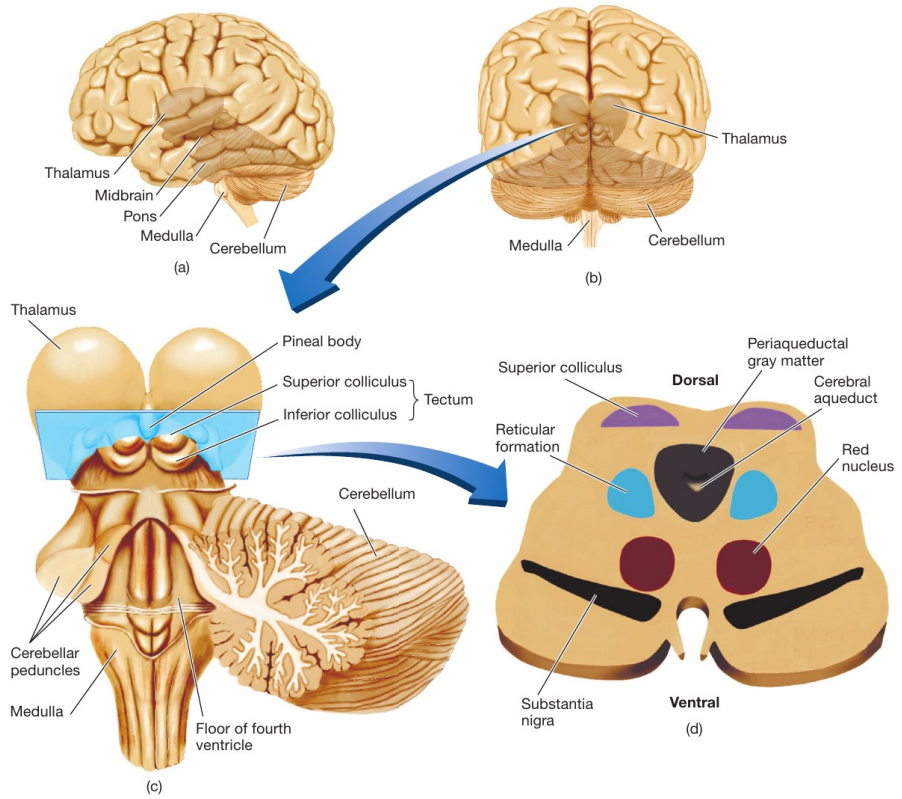
Guo et al. (2021), Video 2. Reach initiation during pontine stimulation.

Cerebellar circuits refine timing and coordination

The **cerebellum** contains a folded cortical sheet and deep nuclei that receive its output. **Purkinje cells** are the major output neurons of the cerebellar cortex. Their axons inhibit targets in deep nuclei and related structures, allowing cortical processing within the cerebellum to shape the signals leaving the system.

Inputs carry information about planned movements, sensory events, and the body's current state. Cerebellar circuits use this information to help coordinate the timing and scale of action. Damage can produce **ataxia**, a disorder of coordination in which movement becomes irregular or poorly directed even when muscles can still generate force.

Cerebellar output travels through several routes to motor and other systems. Some pathways reach cortex through the thalamus, while others influence brainstem nuclei. This distributed arrangement explains why cerebellar contributions can affect reaching, balance, gait, and eye movements instead of controlling only one named muscle group.



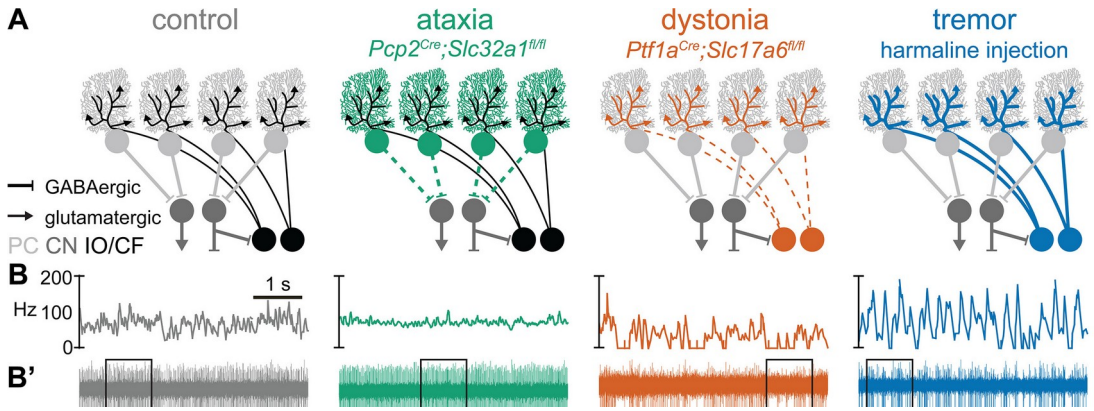
Carlson and Birkett (2022), Fig. 3.16, p. 82.

Different cerebellar disorders produce distinct movements

Van der Heijden and colleagues examined cerebellar nuclear activity in mouse models with ataxia, dystonia, or tremor. **Dystonia** includes abnormal sustained or intermittent muscle contractions, while **tremor** is rhythmic oscillatory movement. These behaviors differ from ataxia, even though all can involve disrupted signaling within cerebellar output circuits.

The investigators found different patterns of spiking in cerebellar nuclei across the models. They then used optogenetic methods to impose selected activity patterns and examine the resulting behavior. The experiment tested whether the temporal structure of output could help produce particular movement abnormalities, beyond a simple increase or decrease in activity.

The accompanying recording shows freely moving control and affected mice from the original study. Their behavior makes the clinical terms distinguishable through actual movement. The study's recordings and manipulations then explain how distinct patterns in a shared anatomical output system can contribute to different observable motor phenotypes.



van der Heijden et al. (2024), Fig. 1A–B. Cerebellar output and recorded spike patterns.



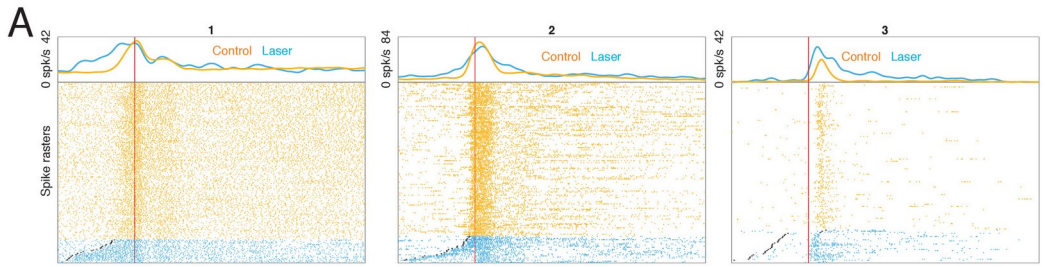
van der Heijden et al. (2024), Video 1. Control and movement-disorder phenotypes.

Cerebellar feedback adjusts the details of a reach

Guo and colleagues recorded activity in motor cortex and deep cerebellar nuclei during trained reaching. They compared control trials with trials in which pontine stimulation perturbed the pathway. Neural responses still accompanied movement, but their differences between conditions contained information about the accompanying differences in hand velocity.

The researchers used the recorded activity to estimate movement-related variables. Such an estimate asks whether a population carries information useful for describing the movement. In this study, the comparison placed altered cortical and cerebellar signals alongside altered behavior during disruption of a known route between those regions.

The experiment separates several levels of explanation. Cortex and cerebellum identify major structures, the pontine relay identifies a path through them, and the changing neural patterns describe activity within that path. The resulting reach is produced by the interacting system, including downstream circuits that turn these signals into coordinated muscle activity.



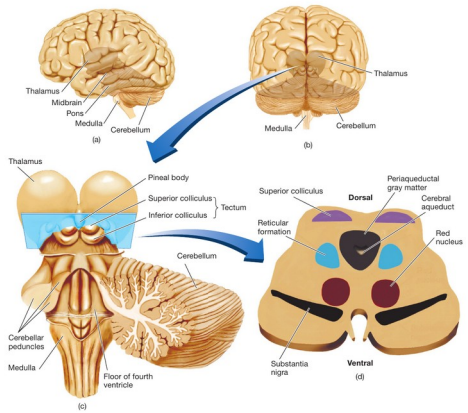
Guo et al. (2021), Fig. 7A. Movement-related cortical population activity.

Medullary circuits regulate breathing and circulation

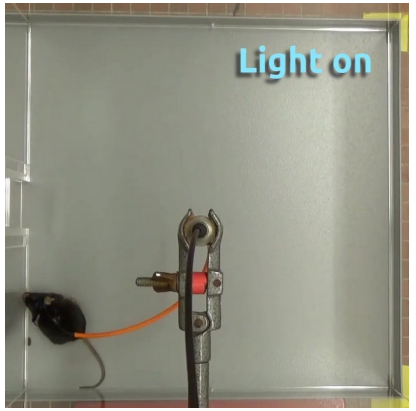
The **medulla oblongata** lies at the lower end of the brainstem, continuous with the spinal cord. It contains populations that regulate breathing, cardiovascular function, and protective reflexes such as swallowing. These circuits operate continually while receiving information about the body's condition and signals descending from other brain regions.

Breathing requires coordinated activation of muscles, while heart-rate regulation changes activity in autonomic pathways reaching the heart. These processes can change together during defense or exercise, but they remain anatomically distinct outputs. A faster breathing rate does not require the heart to change in the same direction under every condition.

The accompanying hypothalamic-to-PAG stimulation recording shows a mouse becoming immobile during the authors' experiment. Separate measurements in that study identified changes in breathing and heart rate. The scene places the bodily response in a behavioral setting, while the medulla and other lower circuits provide routes that implement its internal components.



Carlson and Birkett (2022), Fig. 3.16, p. 82.



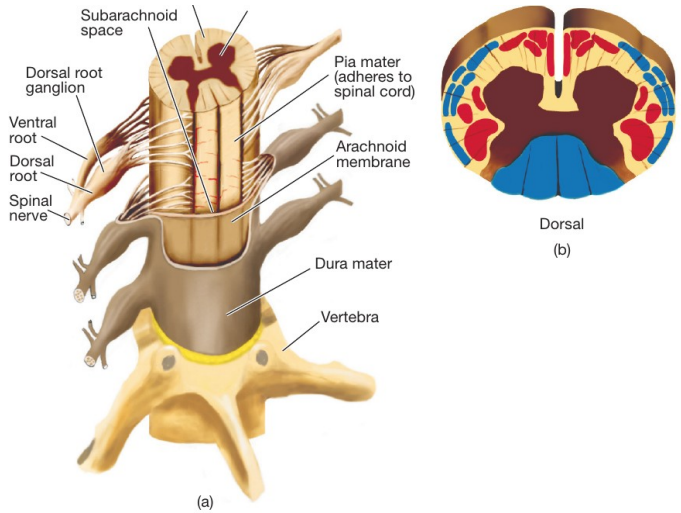
Cola et al. (2024), Video 2. Immobility during pathway stimulation.

The spinal cord carries signals and organizes responses

The **spinal cord** extends through the vertebral canal and communicates with the body through spinal nerves. Its outer white matter contains ascending and descending axon bundles. Its inner gray matter contains neuronal cell bodies and local circuits, producing a different arrangement from the gray sheet surrounding much of the cerebrum.

Ascending pathways carry information toward the brain, while **descending pathways** carry influence toward spinal targets. Local interneurons also process signals within and between spinal segments. This organization allows sensory information to affect movement through short spinal routes as well as through longer routes involving brainstem and cortical structures.

A spinal **interneuron** communicates with other neurons rather than sending its axon directly to skeletal muscle. Interneuronal populations coordinate patterns such as alternating limb activity. The spinal cord therefore participates in organizing action, in addition to carrying messages between the brain and peripheral tissues along its longitudinal pathways.



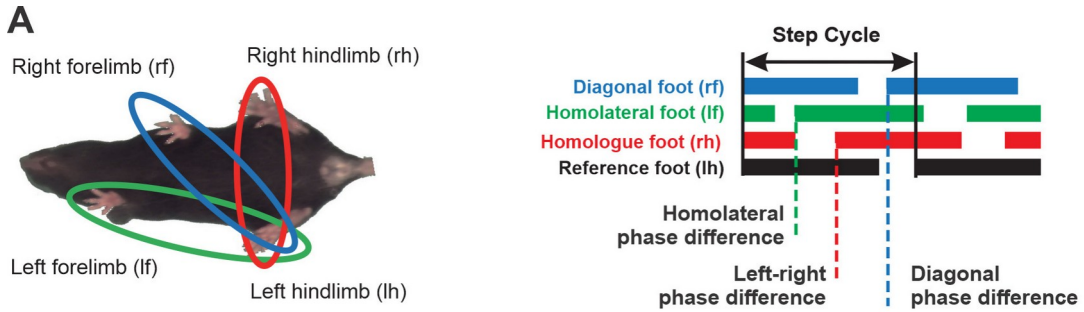
Carlson and Birkett (2022), Fig. 3.18, p. 84.

Spinal V3 neurons contribute to coordination between limbs

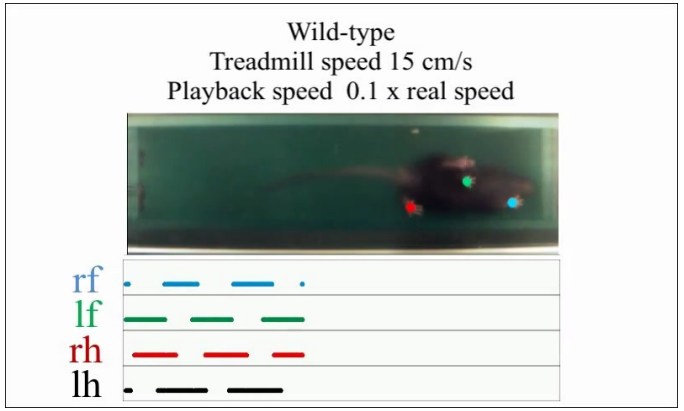
Zhang and colleagues studied **V3 interneurons**, a genetically identifiable spinal population with excitatory outputs. Some of these neurons cross the midline and influence circuits on the opposite side. The investigators compared locomotion in control mice and mice in which transmission from the selected population had been disrupted.

The study examined how limb coordination changed with walking speed. Coordinating the forelimbs and hindlimbs requires timing across several spinal segments, while coordinating left and right requires communication across the midline. V3 populations participate in this distributed organization rather than serving as the final motor neurons that contact limb muscles.

At the lower treadmill speed, control and V3-disrupted animals can both produce repeated steps, but their limb timing differs. The sequence depends on communication between spinal populations serving different limbs. Comparing the phases of foot placement identifies coordination changes that an overall measure of distance traveled would not capture.



Zhang et al. (2022), Fig. 5A. Limb phases within the step cycle.



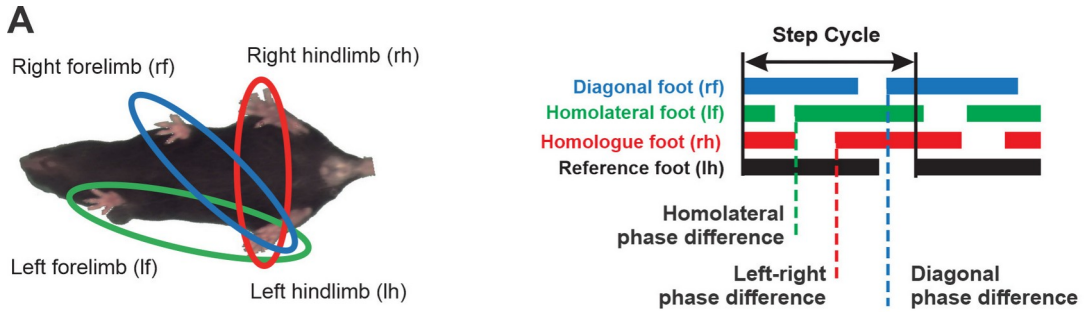
Zhang et al. (2022), Fig. 5, Video 1. Walking at 15 cm/s.

Walking speed changes the demands on spinal coordination

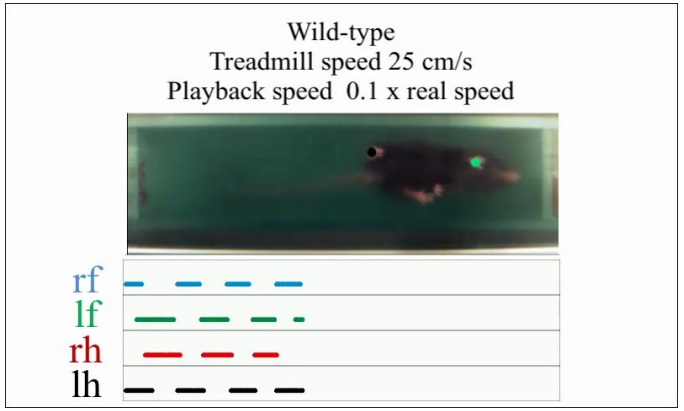
Zhang and colleagues recorded mice walking at several treadmill speeds after disrupting V3 transmission. The limb movements had to remain coordinated while step timing changed. Comparing speeds tested whether the selected spinal population made the same contribution during all locomotion or became especially important under particular movement conditions.

At a faster treadmill speed, each limb must complete its support and swing phases in a shorter interval. The body must remain supported while feet change position. The experiment measured how the order and timing of these events changed when V3 transmission was disrupted, rather than treating faster walking as a larger version of slow walking.

A spinal circuit can contain populations recruited differently as movement changes. The direction of an axon, the segments it reaches, and the cells it excites determine which parts of the pattern it can influence. The study uses these anatomical properties to explain why speed-dependent coordination depends on more than a single rhythm generator.



Zhang et al. (2022), Fig. 5A. Limb phases within the step cycle.



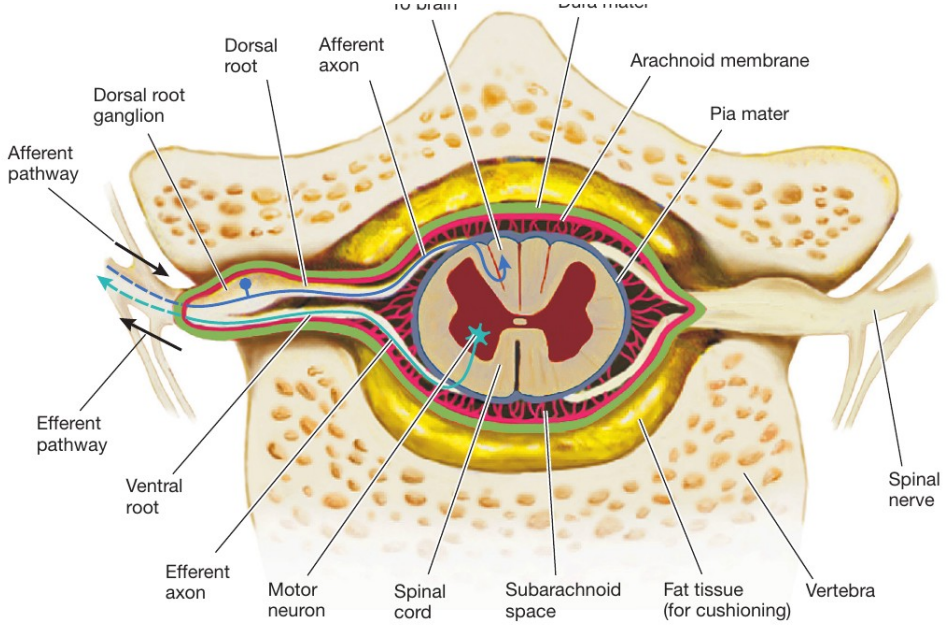
Zhang et al. (2022), Fig. 5, Video 2. Walking at 25 cm/s.

Sensory and motor fibers enter through different roots

A **dorsal root** carries sensory axons into the spinal cord, while a **ventral root** carries motor axons outward. The roots join to form a mixed **spinal nerve**, containing fibers traveling in both functional directions. Their physical union does not cause the signals to lose their separate sensory and motor roles.

Sensory cell bodies lie in the **dorsal root ganglion**, outside the spinal cord. One branch of a sensory neuron's axon travels toward peripheral receptors, while another enters central tissue. Motor-neuron cell bodies lie within the spinal gray matter, and their axons leave through the ventral root toward muscles.

An **afferent** pathway carries information toward a specified central structure, while an **efferent** pathway carries information away from it. These terms describe signal direction relative to the structure being discussed. At the spinal cord, sensory roots are afferent and motor roots are efferent, even when both travel together farther out in a nerve.



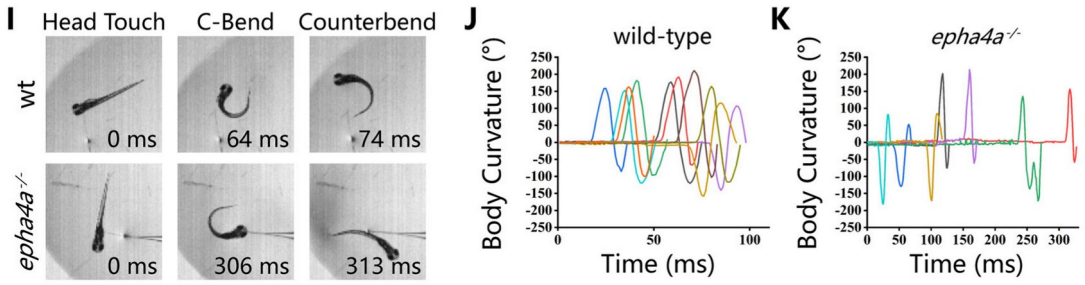
Carlson and Birkett (2022), Fig. 3.20, p. 87.

Guidance signals establish alternating spinal movement

Wang and colleagues examined **Epha4a**, a receptor involved in developmental signaling between cells, in zebrafish. They compared normal larvae with animals carrying a mutation that disrupted this signaling. The study examined spinal circuitry, calcium activity, and behavior to investigate how altered developmental organization affected later left-right movement patterns.

Normal larvae produce a rapid bend after a tactile stimulus and then generate coordinated swimming movements. The original control recording shows this sequence after a touch near the head. The response provides a visible example of sensory input recruiting motor output through an intact developing sensorimotor system in the animal.

The study's mutant comparisons tested how that sequence changed when guidance signaling was disrupted. Developing axons must reach appropriate positions and targets before a circuit can coordinate activity. The behavioral observations therefore place molecular guidance within a larger anatomical problem: building the routes that allow the two sides to act in sequence.



Wang et al. (2025), Fig. 3I–K. Control and mutant startle sequences.



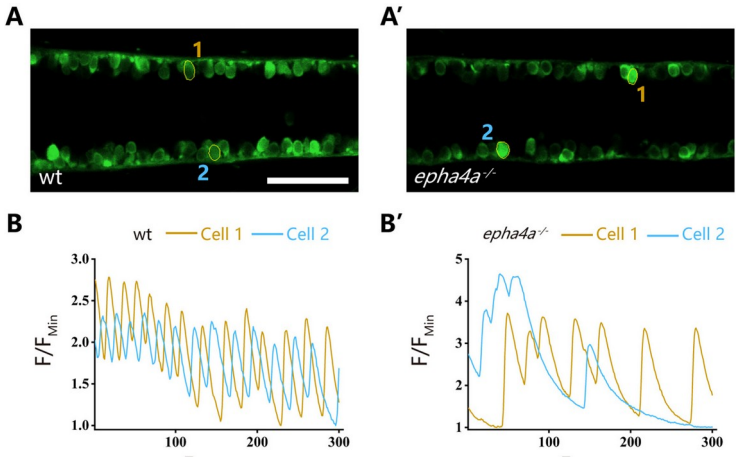
Wang et al. (2025), Video 3. Control larval startle response.

Altered spinal wiring disrupts left-right activity patterns

Wang and colleagues found abnormal left-right movement patterns in zebrafish lacking normal Epha4a function. Calcium recordings from motor-related populations also showed altered activation patterns. Measuring both activity and movement connected the developmental manipulation to changes in the operation of the spinal system, rather than describing body shape alone.

After head touch, normal larvae produce a coordinated bend followed by counterbending and swimming. Epha4a-mutant larvae can produce abnormal sequences under the same type of stimulation. The direction and timing of body curvature provide behavioral measurements of how the developing circuit organizes left-right muscle activity after a defined sensory input.

The nervous system must both generate activity and distribute it appropriately between sides. Synchronous activation can produce a different body movement from alternating activation even when many of the same muscles remain present. Developmental guidance helps establish the spatial arrangement that allows later signals to recruit an organized pattern of contraction.



Wang et al. (2025), Fig. 4A–B′. Bilateral calcium signals in control and mutant larvae.



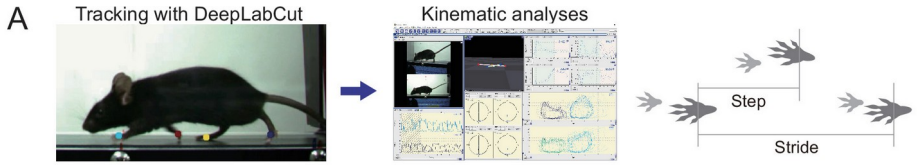
Wang et al. (2025), Video 4. Mutant larval startle response.

Spinal canal-contacting neurons influence locomotion

Nakamura and colleagues developed a method to label and manipulate mouse CSF-contacting neurons around the spinal central canal. They traced their axons and identified contacts with other spinal populations. This work extended the study of the canal's sensory interface beyond larval fish to a mammalian spinal preparation with measurable locomotor behavior.

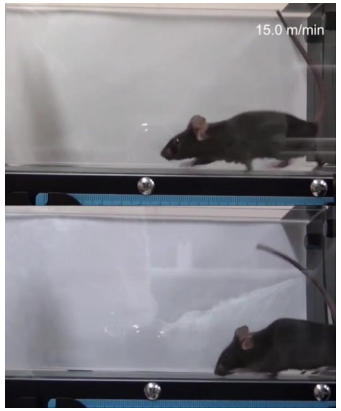
The investigators expressed an inhibitory **designer receptor**, a protein activated by an administered compound, in the selected neurons. Activating that receptor reduced their activity during treadmill tests. The mice showed reduced locomotor speed and step frequency, identifying a contribution of this internal spinal sensory population to the ongoing movement pattern.

The treadmill accelerated during one test, requiring the mice to increase their stepping rate while maintaining coordinated support. Inhibiting CSF-contacting neurons reduced performance under these demands. This result places cells beside the central canal within a functional motor-control network, even though their specialized endings sample an internal compartment rather than receptors in the skin.



Pkd2l1-Cre + AAV-Svn-DIO-hM4Di

Nakamura et al. (2023), Fig. 7A. Treadmill tracking and kinematic analysis.



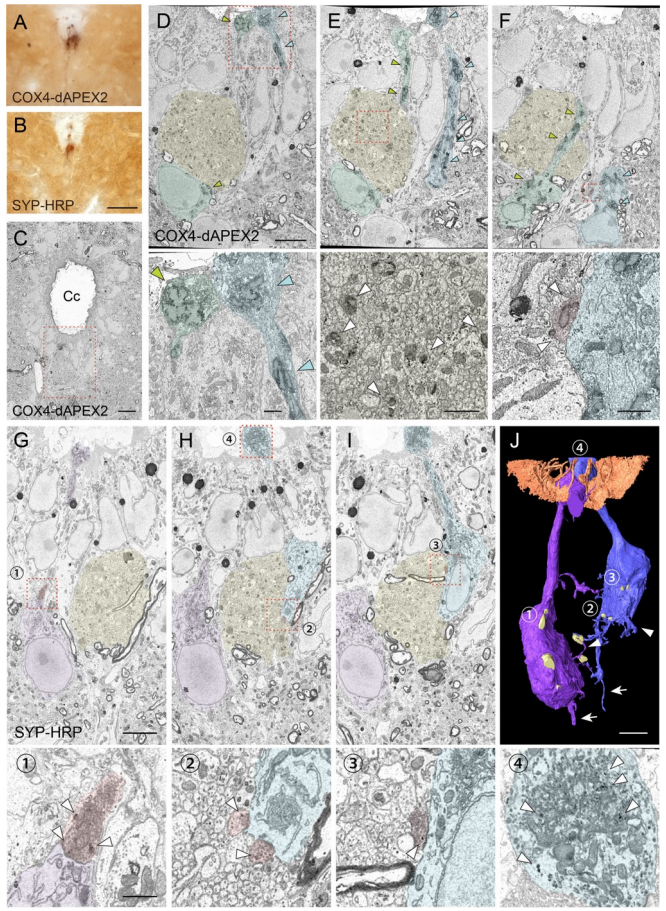
Nakamura et al. (2023), Video 6. Treadmill locomotion during neuronal inhibition.

Tracing reveals an internal spinal sensory network

Nakamura and colleagues used single-cell labeling to follow CSF-contacting neurons and their axons through cleared spinal tissue. Many axons extended rostrally in bundles near the ventral cord. Labeling individual cells separated their paths from the dense surrounding tissue, allowing the direction and targets of the projections to be examined.

Electron microscopy and physiological measurements identified communication among these neurons and with other spinal populations. Some projections reached axial motor neurons and premotor interneurons. These targets provide routes through which information sampled near the central canal can influence neurons that organize posture and the activation of body muscles.

The anatomy and locomotor manipulation answer complementary questions. Tracing establishes where a signal can travel, while reducing activity tests the contribution of the selected cells during behavior. Together, they explain a spinal sensory route whose inputs arise inside the central nervous system and whose outputs reach local motor-control circuits.



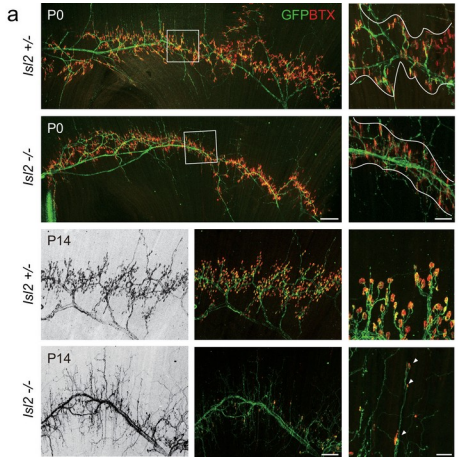
Nakamura et al. (2023), Fig. 4. Electron microscopy of spinal canal-contacting circuits.

Motor pools require precise positions and muscle targets

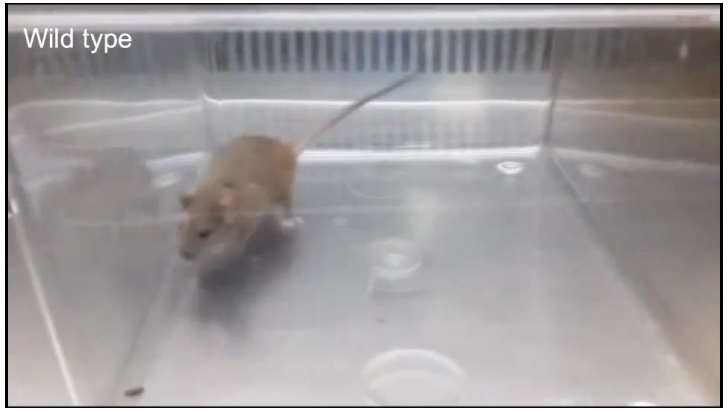
Lee and colleagues studied **Isl2**, a transcription factor that regulates gene expression during motor-neuron development. A **motor pool** is a group of motor neurons supplying a particular muscle. The investigators examined how loss of Isl2 changed the position of these pools and the organization of their peripheral projections.

The mutation dispersed motor-neuron groups and altered terminal branching and neuromuscular organization. Sensory feedback to the affected circuits was also impaired. These findings show that motor development must coordinate cell identity, central position, peripheral targeting, and input from the body, rather than producing motor neurons without specifying their partners.

Adult Isl2-mutant mice showed altered gait alongside the developmental changes in motor pools and their peripheral targets. A misplaced neuron can receive a different set of inputs, and an altered terminal can activate muscle differently. The findings explain movement through the organization of the complete sensorimotor route, from central position to peripheral contact.



Lee et al. (2023), Fig. 8a. Motor axon terminals and muscle endplates.



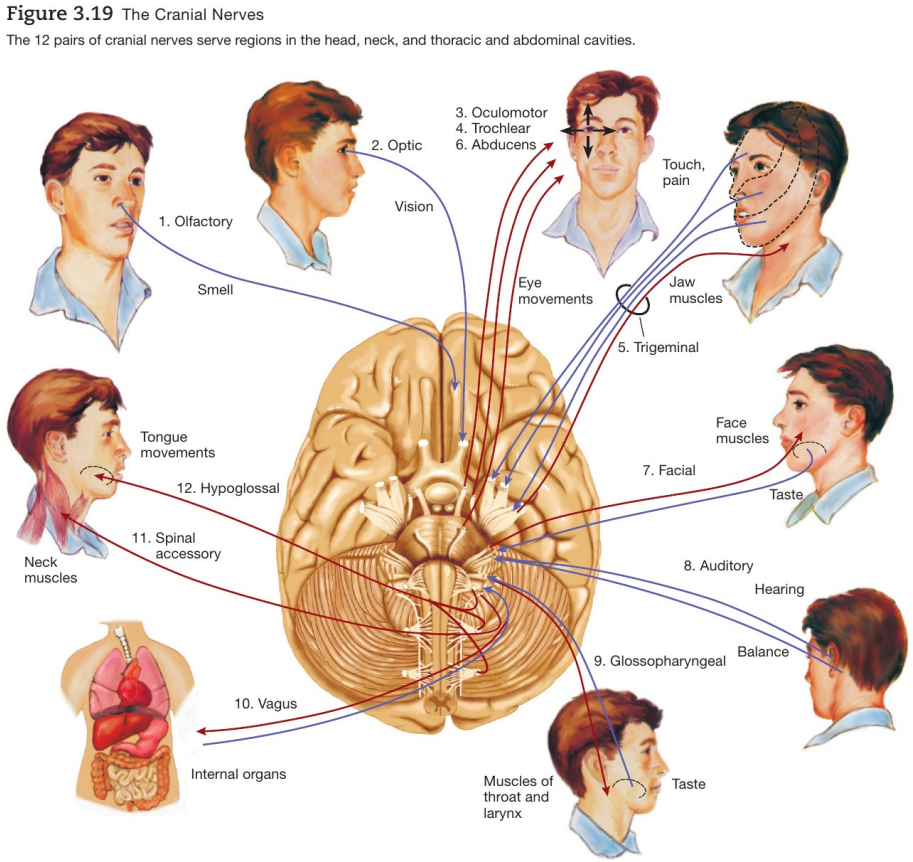
Lee et al. (2023), Video 3. Adult control and Isl2-mutant gait.

Cranial nerves carry head and internal-organ signals

Twelve pairs of **cranial nerves** emerge from the brain or brainstem and serve sensory, motor, or mixed functions. The optic nerve carries visual information, while the oculomotor, trochlear, and abducens nerves control eye muscles. Their distinct routes separate detecting an object from physically moving the eyes toward it.

The trigeminal nerve carries much facial sensation and controls chewing muscles. The facial nerve controls many facial muscles and carries taste information from part of the tongue. The vestibulocochlear nerve carries hearing and balance information. A single named nerve can therefore contain several functional components rather than one uniform message.

The **vagus nerve**, the tenth cranial nerve, extends beyond the head into the chest and abdomen. It contains sensory fibers from internal organs and motor fibers contributing to parasympathetic regulation. This broad distribution provides a major route for communication between the brainstem and organs involved in circulation, breathing, and digestion.



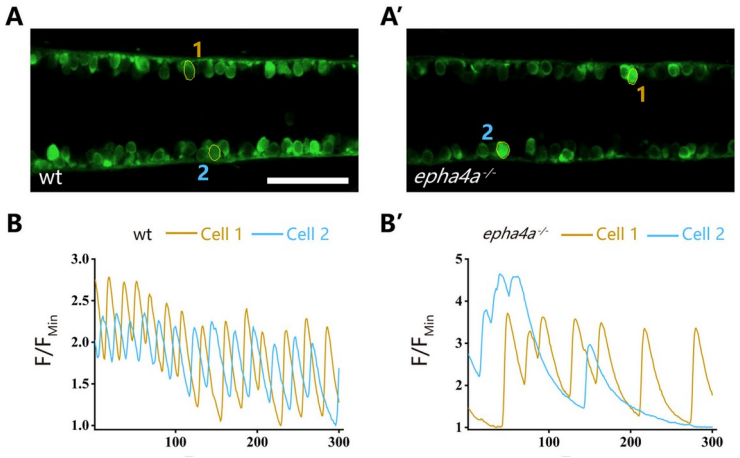
Carlson and Birkett (2022), Fig. 3.19, p. 86.

Calcium imaging exposes the timing of motor activity

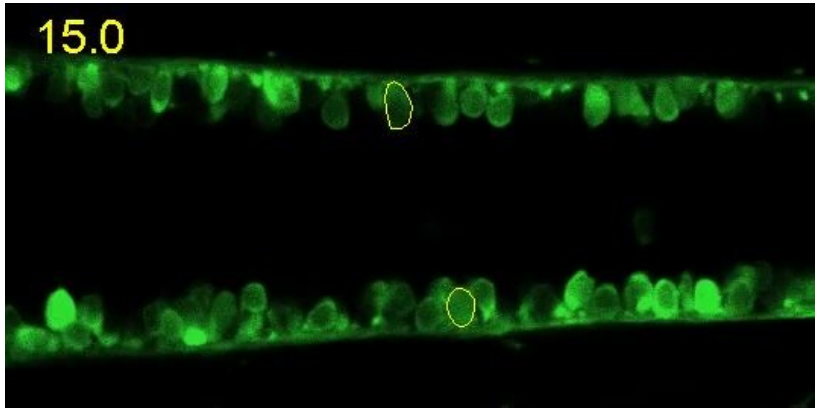
Wang and colleagues recorded calcium signals in developing zebrafish motor-related neurons while studying Epha4a function. The normal recording shows alternating activation across the two sides. This timing is an observable property of a functioning circuit, distinct from a still image showing only where the cells and axons lie.

The study compared normal and disrupted guidance conditions to ask how developmental organization affects that timing. A pathway must place excitation and inhibition at appropriate targets before the network can alternate its output. Changing where neurons project can therefore change the temporal pattern produced later by the same developing spinal system.

Alternating calcium signals identify different times of activation on the two sides of the developing preparation. Those events occur in labeled neuronal populations before their output is expressed as a body movement. Comparing their timing with axon organization helps explain how developmental placement of projections contributes to the later coordination of action.



Wang et al. (2025), Fig. 4A–B′. Bilateral calcium signals in control and mutant larvae.



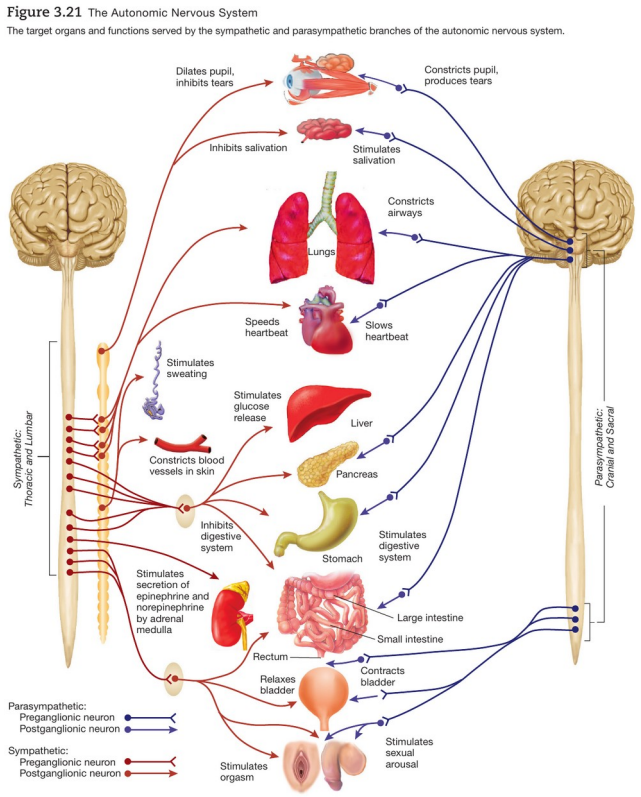
Wang et al. (2025), Video 5. Alternating motor-related calcium signals.

Autonomic pathways use peripheral relays

The **autonomic nervous system** regulates cardiac muscle, smooth muscle, and glands. Most autonomic motor routes use two neurons in sequence: a **preganglionic neuron** in the central nervous system and a **postganglionic neuron** in a peripheral ganglion. The second neuron's axon reaches the target tissue and changes its activity.

This arrangement differs from the somatic motor route to skeletal muscle, where one motor-neuron axon runs from the central nervous system to muscle fibers. An autonomic ganglion creates a peripheral relay. Inputs can be distributed and modified there before the final signal reaches an organ's muscle or secretory cells.

Autonomic regulation operates together with sensory information from internal organs. Stretch, chemical conditions, and other bodily signals reach central circuits through visceral afferents. The returning output can change heart rate, airway tone, digestion, or gland secretion, creating feedback loops that continually adjust organ function to current bodily demands.



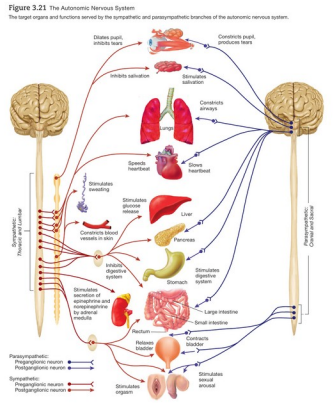
Carlson and Birkett (2022), Fig. 3.21, p. 88.

Sympathetic outflow mobilizes several organ systems

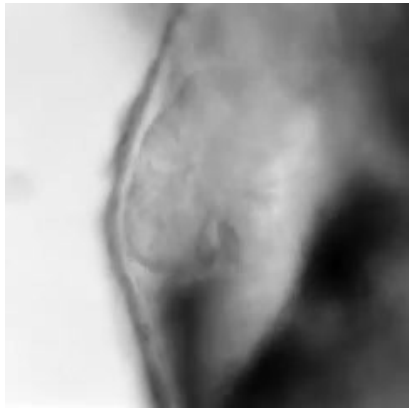
The **sympathetic division** originates mainly from thoracic and upper lumbar spinal regions. Many preganglionic axons reach ganglia near the spinal column, while others reach ganglia closer to abdominal organs. Postganglionic axons then travel to targets, allowing a central signal to influence several structures through peripheral relays.

Sympathetic activity can increase heart rate, alter blood-vessel tone, and change digestive activity. These effects depend on the target tissue and the receptors it expresses. Preganglionic neurons release acetylcholine, while many postganglionic sympathetic neurons release norepinephrine; sweat-gland pathways provide an important exception to that general postganglionic transmitter pattern.

The accompanying recording shows the beating heart of a control zebrafish embryo from a developmental study. Cardiac pacemaker cells generate the rhythm, while autonomic signals later adjust how the heart operates. This embryonic preparation had not yet acquired detectable neuronal innervation, making the scene an example of intrinsic beating rather than sympathetic stimulation.



Carlson and Birkett (2022), Fig. 3.21, p. 88.



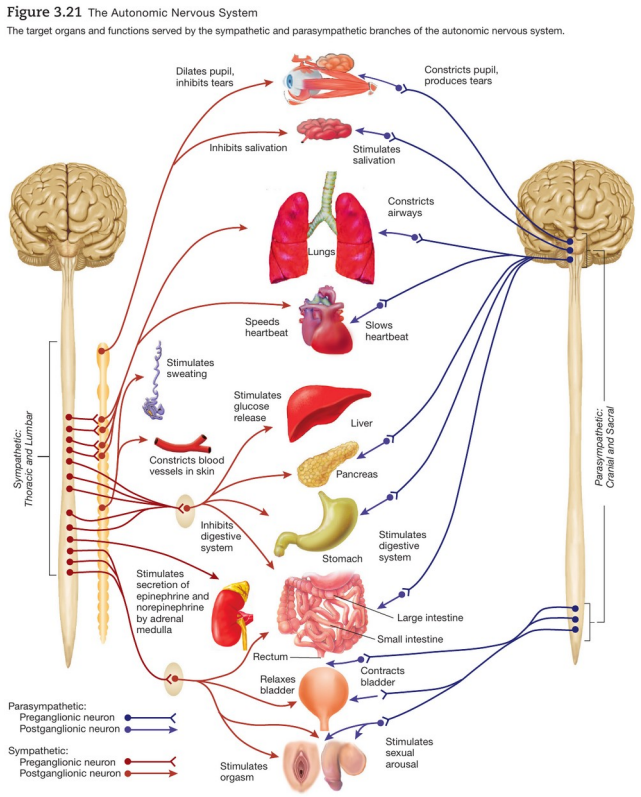
Burkhard et al. (2018), Fig. 6, Video 1. Intrinsic beating of an embryonic heart.

Parasympathetic relays lie near the organs they regulate

The **parasympathetic division** has outflow from selected brainstem nuclei and sacral spinal regions. Its preganglionic axons often travel farther before reaching ganglia near or within their target organs. Many postganglionic axons are correspondingly short, giving this division a different peripheral arrangement from the sympathetic chains beside the spinal column.

Parasympathetic preganglionic neurons release acetylcholine onto nicotinic receptors in the ganglion. Postganglionic neurons also release acetylcholine, usually acting on muscarinic receptors at the organ. The same transmitter can therefore act at two stages while producing different cellular effects because the receiving cells express different receptor types.

Parasympathetic activity can slow the heart and promote several digestive functions, but organ control depends on ongoing interactions among local and central systems. Sympathetic and parasympathetic outputs are not two switches that always turn every organ fully on or off together. Their anatomical targets and receptor mechanisms determine the resulting response.



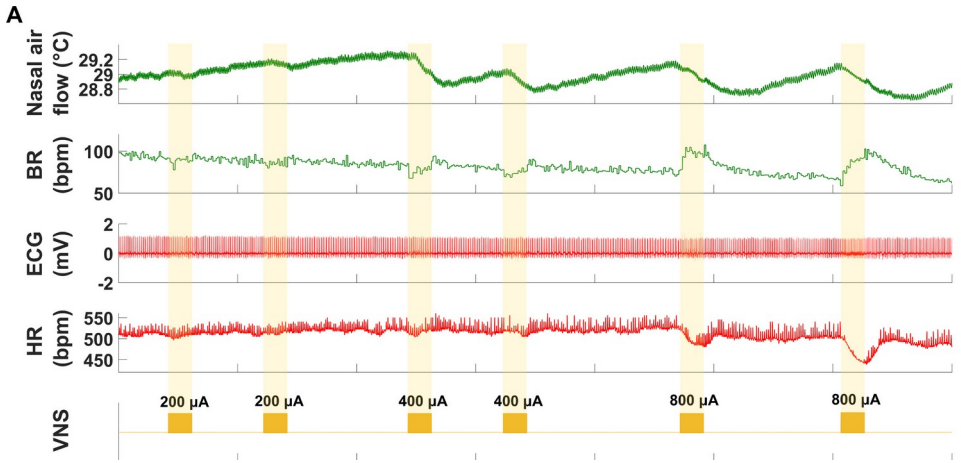
Carlson and Birkett (2022), Fig. 3.21, p. 88.

Vagus stimulation changes heart rate in an awake mouse

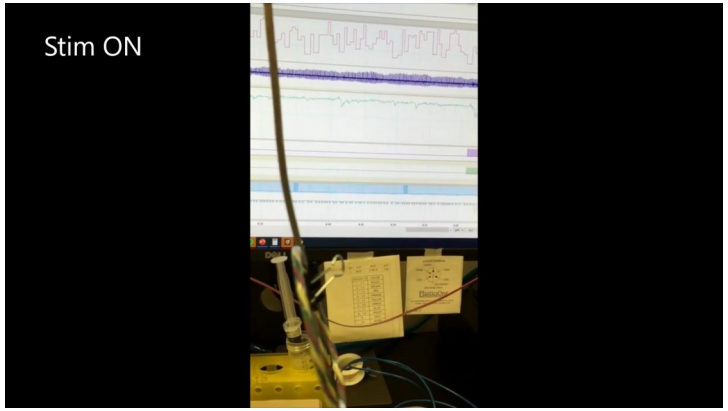
Mughrabi and colleagues developed a long-term cuff electrode around the mouse vagus nerve. The implant allowed the nerve to be stimulated repeatedly while physiological signals were recorded. A **cuff electrode** surrounds a peripheral nerve, providing an electrical interface to its axons without requiring the entire nerve to be cut.

The researchers measured heart-rate responses to stimulation across the period after implantation. The original movie shows an awake mouse with a heart-rate trace and the marked stimulation interval. The change in the recorded cardiac signal demonstrates a physiological response while the animal remains conscious in the chronic experimental preparation.

The cervical vagus contains both sensory and motor fibers, so cuff stimulation can recruit more than one pathway. The study used physiological responses to validate the interface rather than claiming that every activated axon had the same function. The preparation provides an experimental route from a named cranial nerve to measurable organ regulation.



Mughrabi et al. (2021), Fig. 2A. Respiratory cardiac and stimulation traces.



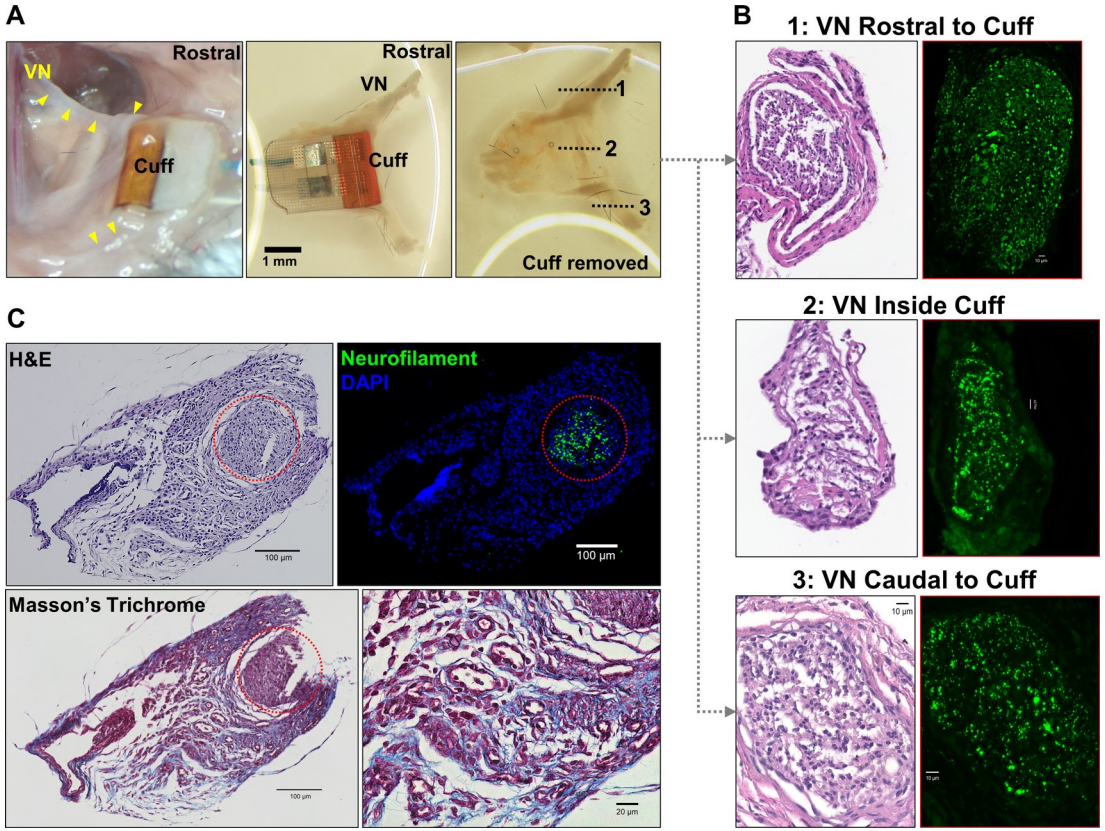
Mughrabi et al. (2021), Video 1. Heart-rate response to vagus stimulation.

A chronic nerve interface must preserve living tissue

Mughrabi and colleagues examined the vagus and surrounding tissue after the electrode had remained implanted. Histological sections showed the nerve within its local tissue environment. **Histology** examines thin tissue sections, allowing researchers to inspect axons, connective tissue, and the material surrounding an implanted device at a much finer scale than behavior alone.

The implanted region developed a fibrotic covering, but the study did not find the axonal pathology expected from major nerve injury in the examined preparations. The investigators also tested vagally mediated physiological responses. These observations addressed whether a usable electrical response was accompanied by preserved structure and ongoing nerve function.

The anatomical interface matters because stimulating a nerve repeatedly requires more than producing one successful pulse. The device must remain positioned beside the intended axons while the surrounding tissue reacts to it. Structural examination and repeated physiological tests therefore establish different aspects of how a chronic peripheral interface operates over time.



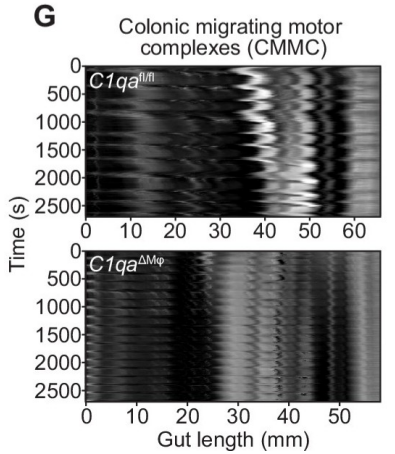
Mughrabi et al. (2021), Fig. 6. Implanted vagus and tissue sections.

The intestinal wall contains circuits that generate movement

The **enteric nervous system** consists of neuronal networks within the gastrointestinal tract. These circuits help coordinate **peristalsis**, the organized contractions that propel material along the intestine. Central autonomic signals influence the gut, but local networks can also organize movement within isolated tissue when the relevant cells and connections remain intact.

Pendse and colleagues studied intestinal macrophages, immune cells positioned near enteric neurons. They identified a macrophage product called **C1q** and tested how removing it from those cells affected intestinal function. The study placed immune regulation within the local tissue environment surrounding a peripheral neural network involved in organ movement.

Isolated colonic tissue continues to produce organized contractions when its local neural and muscle systems remain viable. This preparation separates local movement generation from ongoing signals descending from the brain. Altering macrophage C1q within the tissue environment then tests how neighboring cells regulate the operation of a peripheral neural network.



Pendse et al. (2023), Fig. 6G. Colonic contraction maps.



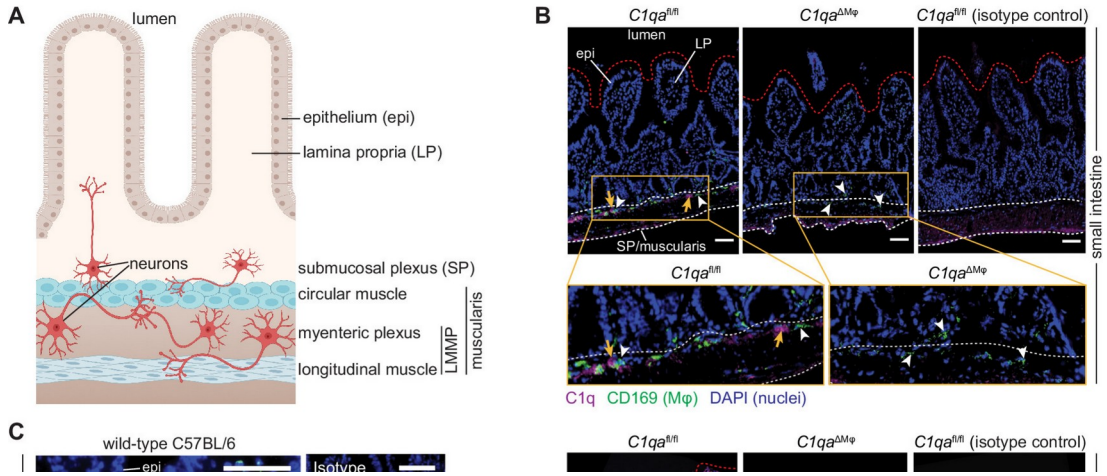
Pendse et al. (2023), Fig. 6, Video 1. Colonic peristalsis in the control preparation.

Nerve-adjacent macrophages influence intestinal motility

Pendse and colleagues found C1q-producing macrophages near neuronal networks in the intestinal wall. The **myenteric plexus** is a major enteric network positioned between muscle layers. Imaging these locations placed the immune cells near the neurons that help coordinate contraction, giving a specific anatomical setting for testing their influence on gut function.

Deleting C1q from macrophages changed enteric neuronal gene expression and increased neurogenic activity of peristalsis. Intestinal transit also became faster in the tested animals. These measurements describe different levels of the response: molecular changes in neurons, altered movement within the tissue, and changed passage of material through the gut.

The experiment expands the cellular account of peripheral control. Enteric neurons act within a living wall containing muscle, glia, immune cells, and other cell types. A product from neighboring macrophages can alter the network's operation, so explaining organ movement requires identifying both the neural circuit and the tissue environment around it.



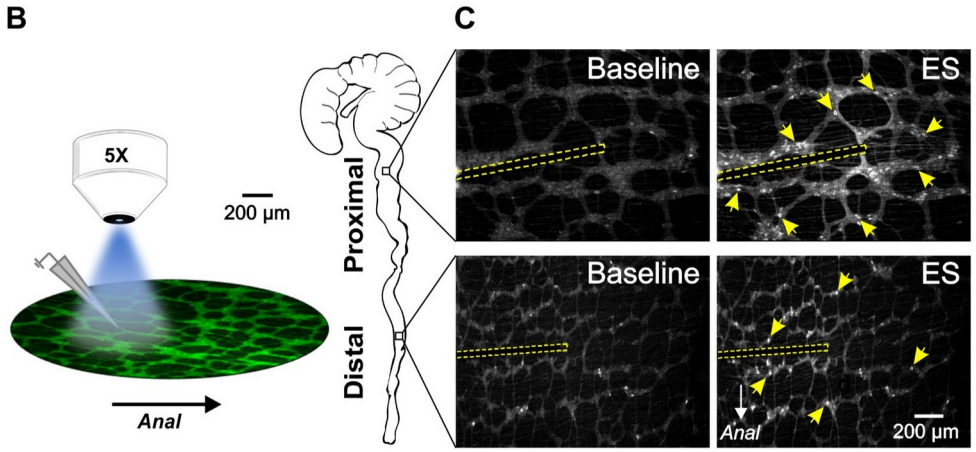
Pendse et al. (2023), Fig. 4A–B. Macrophages beside enteric neural tissue.

Focal stimulation recruits a distributed enteric network

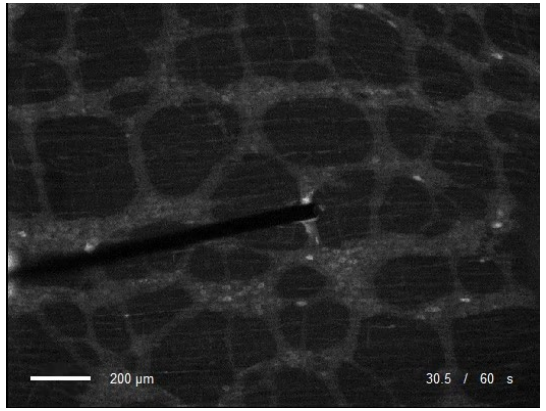
Li and colleagues imaged calcium activity in mouse enteric neurons while stimulating a small fiber strand. The preparation preserved the network within the intestinal wall. Stimulation recruited neurons at several locations, showing how a local input can spread through a peripheral circuit instead of remaining confined to cells directly beside the electrode.

In the proximal colon, focal stimulation produced calcium responses in neurons distributed across the surrounding network. The stimulus entered through a selected fiber strand, while the responding cells occupied several positions. Their spatial distribution allowed the investigators to measure how one input spread through a region capable of producing several intestinal movement patterns.

The investigators combined activity imaging with tracing and examination of neuronal contacts. These methods tested different parts of the same circuit explanation. Calcium imaging showed which cells responded, while anatomical methods examined the routes available between them, connecting the pattern of recruitment with the organization of the enteric network.



Li et al. (2019), Fig. 1B–C. Enteric preparation and activity after stimulation.



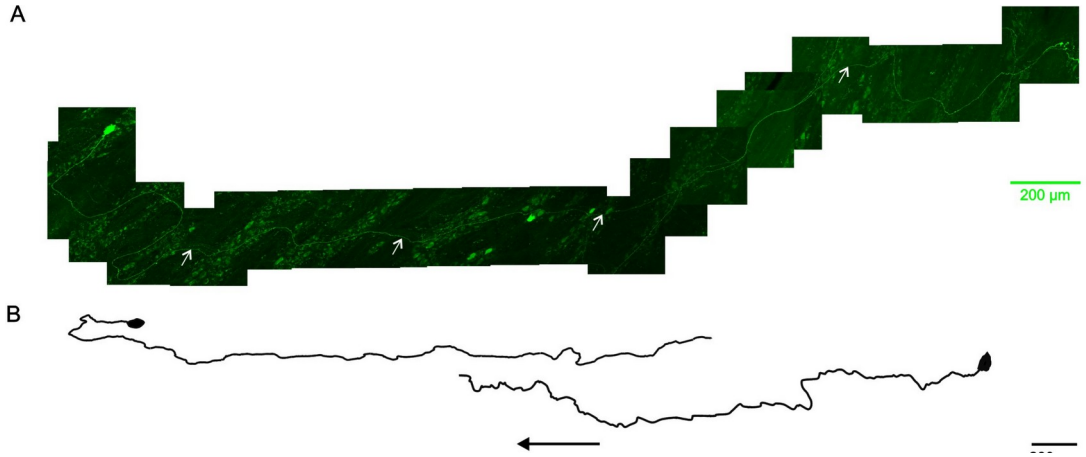
Li et al. (2019), Fig. 1, Video 1. Proximal-colon neuronal responses.

Different intestinal regions contain different wiring patterns

Li and colleagues compared the **proximal colon**, nearer the beginning of the large intestine, with the **distal colon**, nearer its end. These regions produce different combinations of movement patterns. The investigators asked whether their neuronal wiring differed in ways that could account for those regional differences in organ behavior.

The study examined neuronal density, projection length and direction, and responses to local stimulation. The proximal network contained additional organizational features alongside its broader movement repertoire. The comparison therefore treated the enteric nervous system as regionally specialized, rather than assuming the same local circuit repeats unchanged along the entire intestinal wall.

The principle also applies to central anatomy: a region's name gives its position, while its cell populations and targets explain its available operations. In the colon, measuring projection patterns and activity makes that distinction concrete within a peripheral tissue that can generate coordinated movement without a skeletal motor-neuron pathway.



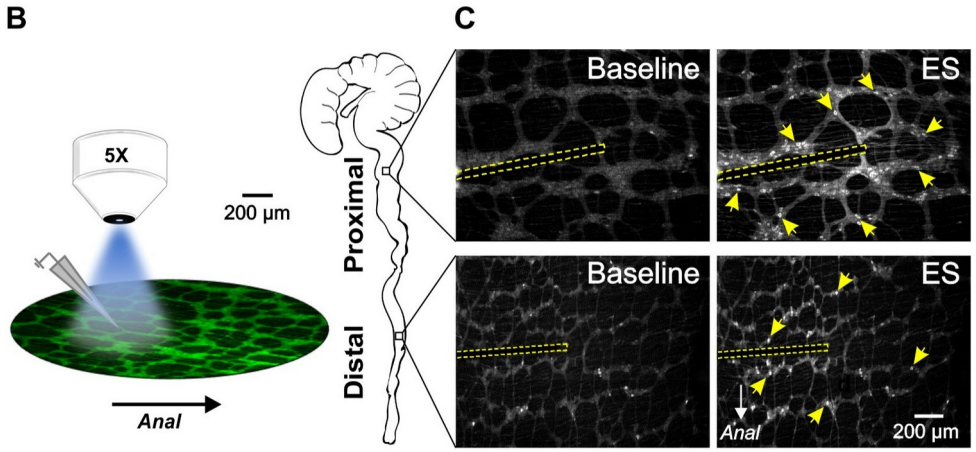
Li et al. (2019), Fig. 7A–B. Enteric axon paths.

Local and central circuits cooperate to regulate the body

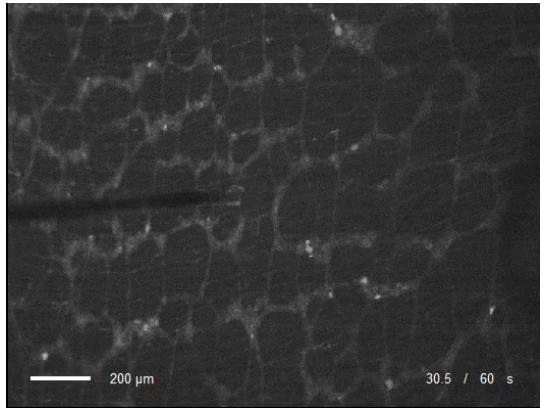
In the distal colon, a separate focal input recruited calcium responses across the local enteric network. Comparing this recruitment with the proximal region tested how regional wiring affects the spread of activity. The experiment places a functional response within a specific peripheral circuit, where local neurons organize an observable pattern of organ movement.

Central autonomic pathways can influence enteric networks, but those inputs act on tissue that already contains sensory cells, interneurons, motor neurons, and muscle. An organ's response therefore emerges from communication across levels. A brainstem signal changes local processing, and local circuits turn that influence into coordinated changes in the intestinal wall.

Across the nervous system, structure specifies where signals can enter, where they can travel, and which cells receive them. Sensory pathways carry information inward, central and peripheral networks transform it, and motor pathways alter muscles or glands. Explaining behavior requires following those actual routes from a defined input to an observable action.



Li et al. (2019), Fig. 1B–C. Enteric preparation and activity after stimulation.



Li et al. (2019), Fig. 1, Video 3. Distal-colon neuronal responses.

Key takeaways

Anatomical directions identify position. Distinguish rostral from caudal, medial from lateral, and ipsilateral from contralateral. Interpret each direction relative to the neuraxis and viewing plane.

Compartments contain active interfaces. Meninges, ventricles, and the spinal central canal define fluid spaces where cilia, fibers, and sensory endings interact.

A region contains several circuits. Identify the selected cells, their input, their axonal target, and the manipulation before explaining a behavioral finding.

Motor control crosses several levels. Cortical, cerebellar, brainstem, and spinal routes cooperate to coordinate action. Local spinal networks contribute to the movement pattern.

Roots separate sensory and motor traffic. Dorsal roots carry afferent fibers, ventral roots carry efferent fibers, and mixed peripheral nerves carry both functional directions.

Organ control combines central and local pathways. Sympathetic and parasympathetic routes use peripheral ganglia. Enteric networks also process information within the intestinal wall and coordinate local movement.